

INTERNAL DOCUMENTATION

Lake Group Website Developer Guide

Static Multi-Page Corporate Site — Architecture, Systems & Reference

Version aligned with repository · July 2026

Production URL: <https://www.lakeoilgroup.com/>

29 HTML pages · 28 live + dashboard demo · Firebase Hosting · PWA-enabled

Contents Guide — How to Find What You Need

Use this page as a roadmap. Each row links to the chapter that explains **what** the technology is, **where** it lives in the repo, **why** it was chosen, and **how** it works.

Tip: Open this PDF in a browser and click any blue link to jump straight to that section. In a PDF viewer, use the bookmark/outline panel or search (Ctrl+F) for filenames like `sw.js`, `flagship.css`, or page names like `fuel.html`.

Find by Technology

Looking for...	Key files	Used on	Section
PWA / offline / install	manifest.webmanifest, sw.js, assets/pwa.js	All pages	§9
SEO / Google / crawlers	robots.txt, sitemap.xml, JSON-LD in HTML	All public pages	§15, §33
Languages (EN / FR / SW)	assets/i18n.js, i18n- content.js	All except 404/offline	§10, §22
Homepage 3D globe	assets/hero-3d.js, hero- 3d.bundle.js	index.html only	§8, §8b
Flagship design / layout	assets/flagship.css, flagship-motion.js	27 interior pages	§7, §7b
Homepage styles / motion	assets/theme.css, motion.js	index.html only	§7
Chat / assistant	assistant.js, assistant- kb.js, FlexSearch	All with chat widget	§11, §11b
News articles	news.js, news-data.js	news.html, news- article.html	§12
Africa operations map	africa-network-map.js, Leaflet	africa- network.html	§13
Shared nav / footer	scripts/templates/, site.js	All pages	§4
Build & deploy	scripts/build_*, package.json	Developer workflow	§14, §16, §40
Firebase hosting	package.json, firebase config	Production deploy	§20
Accessibility	CSS reduced-motion, ARIA, 3D fallback	Site-wide	§17
Every HTML page	*.html (29 files)	Per-page deep dive	§5, §23, §39
Every JS module	assets/*.js	Module reference	§6, §24
All production files	Full repo tree	File inventory	§3

- `manifest.webmanifest` — install metadata, icons, theme colors
- `assets/pwa.js` — SW registration, update toast, `SKIP_WAITING`
- `sw.js` — four caches (v10), precache list, fetch routing
- Cache strategies: network-first, cache-first, stale-while-revalidate, LRU trim
- `offline.html` / `404.html` — fallback pages

SEO — subsection map (§15, §33)

- `robots.txt` — allow/disallow rules, sitemap URL
- `sitemap.xml` — 26 URLs, priorities, lastmod
- Per-page meta: description, Open Graph, Twitter cards, canonical
- `scripts/add_seo_tags.js` — automated injector
- Schema.org JSON-LD by page (Organization, Service, BreadcrumbList, ContactPage)

Pages by category (§5)

- Homepage — `index.html` (`theme.css`, `hero-3d`, `motion.js`)
- Services hub + 7 division pages (fuel, lpg, steel, concrete, logistics, lubricants, container-services)
- Company pages (about, history, leadership, csr, careers, contact, investors, projects, sustainability)
- Network (africa-network, station-locator, fleet, services)
- Media (news, news-article, gallery, media-center, our-story)
- Utility (404, offline, dashboard demo)

Script load order (standard page)

```
i18n-content.js → i18n.js → site.js → flexsearch → assistant-kb.js
→ assistant.js → pwa.js → flagship-motion.js
(Homepage replaces flagship pair with theme.css + motion.js, adds lazy hero-3d.bundle.js)
```

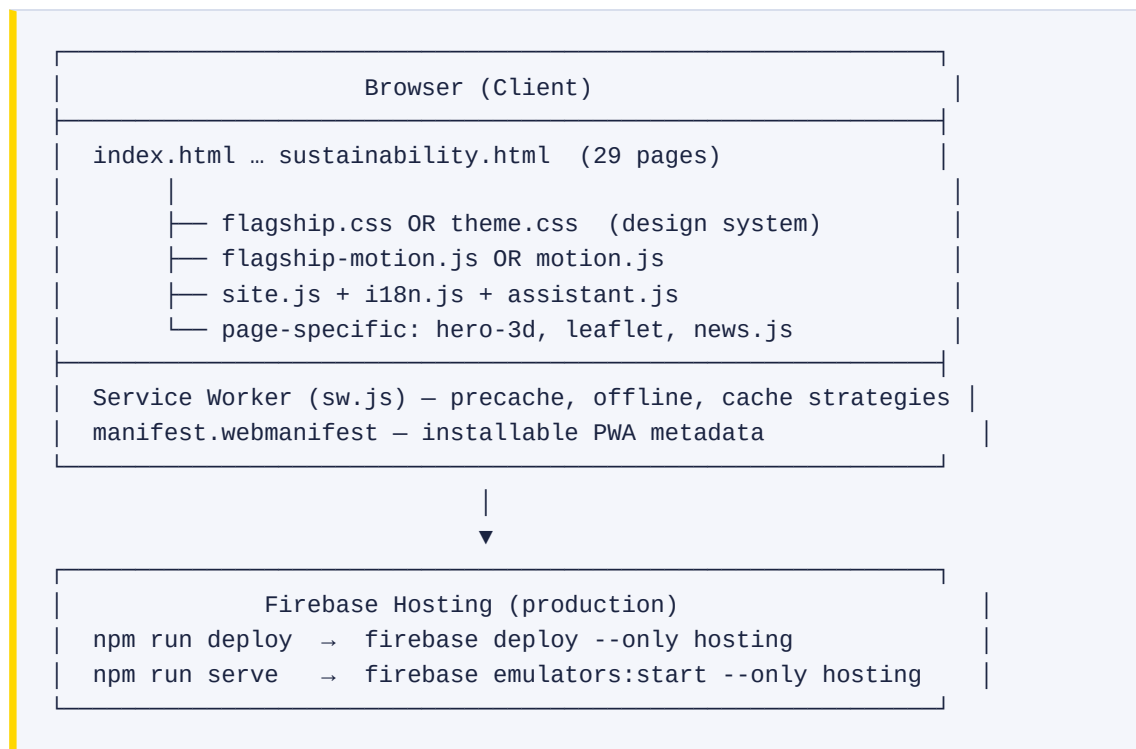
Full Table of Contents

- | | |
|---------------------------------------|-----------------------------------|
| 1. Architecture Overview | 19. Troubleshooting |
| 2. Technology Stack (with rationale) | 20. Firebase Hosting |
| 3. Production File Tree | 21. Image Assets |
| 4. Shared Page Anatomy | 22. i18n Key Conventions |
| 5. HTML Page Reference (All 29 Pages) | 23. Extended Page Docs |
| 6. JavaScript Modules | 24. Extended JS Docs |
| 7. CSS Design Systems | 25. Developer Workflows |
| 7b. Meridian Component Catalog | 26. Security Notes |
| 8. 3D Hero (hero-3d.js) | 27. Performance |
| 8b. 3D Hero Deep Dive | 28. Version History |
| 9. Progressive Web App | 29. HTML Page Index Table |
| 10. Internationalization (i18n) | 30. Asset Index Table |
| 11. Knowledge Assistant | 31. Vendor Libraries |
| 11b. Assistant Deep Dive | 32. Glossary |
| 12. News System | 33. Structured Data by Page |
| 13. Africa Network Map | 34. Cache Strategy Matrix |
| 14. Build & Maintenance Scripts | 35. ADR Summary |
| 15. SEO & Structured Data | 36. Cross-Reference Index |
| 16. Deployment Checklist | 37. Changelog |
| 17. Accessibility | 38. Credits & Licenses |
| 18. Appendices | 39. Page-by-Page Technical Essays |
| | 40. Build Scripts Encyclopedia |

1. Architecture Overview

The Lake Group corporate website is a **static multi-page application (MPA)** — no React, Vue, or Angular on the live site. Each route is a standalone `.html` file at the repository root, served as static files with shared assets in `assets/`.

1.1 High-Level Diagram



1.2 Design Principles

- **Static-first:** No server-side rendering or API dependency for page content. All copy ships in HTML + `i18n-content.js`.
- **file:// safe:** Critical data (translations, geo JSON, assistant KB, 3D bundle) loads via `<script>` globals, not `fetch()`, so double-clicking `index.html` still works.
- **Progressive enhancement:** Pages render without JS; motion, assistant, 3D, and map layers enhance when available.
- **Template-driven chrome:** Nav, mobile nav, footer, and chat widget are generated from `scripts/templates/` via `normalize_nav.py`.

1.3 What Is NOT Live

lake-3d/ is a separate Next.js scroll-driven 3D experience (port 3001). It is *not* linked from any production HTML page. The live homepage 3D is assets/hero-3d.js (bundled), embedded in index.html only.

1.4 Page Load Lifecycle

1. Browser parses HTML; inline critical CSS paints first paint.
2. External CSS (flagship/theme + assistant) loads.
3. Deferred scripts execute in order: i18n-content → i18n → site → flexsearch → assistant-kb → assistant → [page-specific] → pwa → motion.
4. DOMContentLoaded: site.js inits nav, reveals, counters; Lakel18n applies stored language.
5. load: pwa.js registers service worker.
6. index.html only: IntersectionObserver lazy-loads hero-3d.bundle.js when #fuel-experience nears viewport.

1.5 Dual Design System Migration

The site is mid-migration from theme.css + motion.js (legacy) to flagship.css + flagship-motion.js (Meridian). Only index.html still uses the legacy pair. Both remain precached in sw.js until rollout completes. See Section 7.

2. Technology Stack

Complete inventory of production technologies with rationale.

Technology	Role	Why Chosen
HTML5	Page structure	Zero build step for pages; SEO-friendly; works everywhere
CSS3 (inline + flagship/theme)	Styling	Critical CSS inlined per page for FCP; shared system CSS for consistency
Vanilla JavaScript (ES5/IIFE)	Behaviour	No framework runtime; small payload; file:// compatible when bundled
Three.js (vendor)	3D hero	Industry-standard WebGL; bundled for non-module contexts
Leaflet 1.x (vendor)	Africa map	Lightweight maps; Esri/OSM tile support
FlexSearch (vendor)	Assistant search	Client-side full-text index; offline capable
Firebase Hosting	Deployment	CDN, HTTPS, SPA-style rewrites optional; team already uses Firebase CLI
Service Worker	PWA / offline	Precache shell; stale-while-revalidate for assets
Python 3	Build tooling	i18n extraction, font self-hosting, nav normalization
Node.js	KB build, SW lang, QA	build_assistant_kb.js, add_* tag scripts
esbuild (via build_hero_bundle.sh)	3D bundle	Inlines Three.js import into classic script
Self-hosted fonts	Typography	Inter, Bebas Neue, Playfair, Material Symbols — no Google Fonts CDN

3. Production File Tree

Excludes .git, node_modules, .shots, and Chrome QA profiles under scripts/_chrome_profile*.

- └─ 404.html
- └─ FLAGSHIP_DESIGN.md
- └─ QA_REPORT.md
- └─ README.md
- └─ about.html
- └─ africa-network.html
- └─ careers.html
- └─ concrete.html
- └─ contact.html
- └─ container-services.html
- └─ csr.html
- └─ dashboard.html
- └─ favicon.ico
- └─ fleet.html
- └─ fuel.html
- └─ gallery.html
- └─ history.html
- └─ index.html
- └─ investors.html
- └─ leadership.html
- └─ logistics.html
- └─ lpg.html
- └─ lubricants.html
- └─ manifest.webmanifest
- └─ media-center.html
- └─ news-article.html
- └─ news.html
- └─ offline.html
- └─ our-story.html
- └─ package.json
- └─ projects.html
- └─ robots.txt
- └─ services.html
- └─ sitemap.xml
- └─ station-locator.html
- └─ steel.html
- └─ sustainability.html
- └─ sw.js
- └─ assets/
 - └─ africa-map.svg
 - └─ africa-network-map.js
 - └─ assistant-kb.js
 - └─ assistant.css
 - └─ assistant.js
 - └─ data_countries_africa.js
 - └─ flagship-motion.js
 - └─ flagship.css
 - └─ hero-3d.bundle.js
 - └─ hero-3d.js
 - └─ i18n-content.js
 - └─ i18n-content.json
 - └─ i18n.js
 - └─ motion.js
 - └─ news-data.js
 - └─ news.js

```
├─ pwa.js
├─ site.js
├─ theme.css
├─ fonts/
│   ├── fonts.css
│   └─ files/
│       ├── bebas-neue-latin-400-normal.woff
│       ├── bebas-neue-latin-400-normal.woff2
│       ├── inter-latin-300-normal.woff
│       ├── inter-latin-300-normal.woff2
│       ├── inter-latin-400-normal.woff
│       ├── inter-latin-400-normal.woff2
│       ├── inter-latin-500-normal.woff
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│       ├── playfair-display-latin-400-italic.woff
│       ├── playfair-display-latin-400-italic.woff2
│       ├── playfair-display-latin-400-normal.woff
│       ├── playfair-display-latin-400-normal.woff2
│       ├── playfair-display-latin-500-normal.woff
│       ├── playfair-display-latin-500-normal.woff2
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│   ├── diversity_3.svg
│   ├── domain.svg
│   ├── eco.svg
│   ├── forest.svg
│   ├── groups.svg
│   ├── health_and_safety.svg
│   ├── menu_book.svg
│   ├── propane_tank.svg
│   ├── public.svg
│   ├── school.svg
│   ├── shield.svg
│   ├── support_agent.svg
│   ├── trending_up.svg
│   ├── workspace_premium.svg
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│       ├── icon-512.png
│       ├── icon-maskable-192.png
│       └─ icon-maskable-512.png
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│   ├── og-cover.jpg
│   └─ aficd/
│       └─ TA/
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  └─ LakeGas.jpg
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  └─ LakeTrans.jpg
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 - |─ ally-composite-launch.jpg
 - |─ ally-morocco-mou.jpg
 - |─ ally-sino-trucks.jpg
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 - |─ ceo.jpg
 - |─ group-event-2018-2.jpg
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 - └─ photo_1.jpg
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- └─ 9/
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 - └─ photo_2.jpg
 - └─ photo_3.jpg
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 - └─ earth_color_2048.jpg
 - └─ earth_normal_1024.jpg

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│   ├── photo_10.jpg
│   ├── photo_2.jpg
│   ├── photo_3.jpg
│   ├── photo_4.jpg
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│   ├── photo_6.jpg
│   ├── photo_7.jpg
│   ├── photo_8.jpg
│   └─ photo_9.jpg
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│   ├── three.module.min.js
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│   ├── AGENTS.md
│   ├── CLAUDE.md
│   ├── README.md
│   ├── eslint.config.mjs
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│   ├── next.config.ts
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│   ├── tsconfig.json
│   ├── assets/
│   │   └─ lake-group-logo-source.png
│   ├── public/
│   │   ├── favicon.png
│   │   └─ lake-group-logo.png
│   ├── scripts/
│   │   ├── clean-dev.mjs
│   │   └─ prepare_assets.py
├─ src/
│   ├── app/
│   │   ├── favicon.ico
│   │   ├── globals.css
│   │   ├── layout.tsx
│   │   └─ page.tsx
```



```
├─ components/
│   ├─ LandingExperience.tsx
│   └─ scene/
│       ├─ CameraRig.tsx
│       ├─ Droplets.tsx
│       ├─ FloatingParticles.tsx
│       ├─ HeroDrop.tsx
│       ├─ InletConnection.tsx
│       ├─ PetrolStream.tsx
│       ├─ PostEffects.tsx
│       ├─ SceneCanvas.tsx
│       ├─ SceneLights.tsx
│       ├─ StorageTank.tsx
│       ├─ StreamSparkles.tsx
│       ├─ TankFullEffects.tsx
│       └─ VolumetricFog.tsx
│   └─ ui/
│       ├─ AnimatedBackground.tsx
│       ├─ EndingOverlay.tsx
│       ├─ FilmGrain.tsx
│       ├─ HeroOverlay.tsx
│       ├─ LoadingOverlay.tsx
│       ├─ ReplayButton.tsx
│       └─ ScrollIndicator.tsx
├─ context/
│   ├─ PerformanceContext.tsx
│   └─ ScrollContext.tsx
├─ hooks/
│   ├─ useAutoScroll.ts
│   ├─ usePerformanceTier.ts
│   └─ useScrollTrigger.ts
├─ lib/
│   ├─ createEnvMap.ts
│   ├─ curveUtils.ts
│   ├─ randomUtils.ts
│   ├─ suppressThreeDeprecations.ts
│   └─ tankConstants.ts
├─ lake-story-assets/
│   ├─ scene1.jpg
│   ├─ scene2.jpg
│   ├─ scene3.jpg
│   ├─ scene4.jpg
│   ├─ scene5.jpg
│   ├─ scene6.jpg
│   ├─ scene7.jpg
│   └─ scene8.jpg
└─ scripts/
    ├─ _add_ose_keys.js
    ├─ _apply_audit_fixes.js
    ├─ _assistant_notes.md
    ├─ _assistant_qa.js
    ├─ _coverage.json
    ├─ _extracted_en.json
    ├─ _globe_qa.js
    ├─ _globe_qa2.js
    └─ _globe_qa_notes.md
```

- |— _master_en.json
- |— _qa_check_i18n.js
- |— _qa_check_links.js
- |— _qa_i18n_orphans.txt
- |— _qa_matrix.js
- |— _qa_out_i18n.txt
- |— _qa_out_links.txt
- |— _qa_out_matrix.txt
- |— _qa_overflow_check.js
- |— _sw_out_1.json
- |— _sw_out_2.json
- |— _sw_out_3.json
- |— _sw_out_4.json
- |— _sw_src_1.json
- |— _sw_src_2.json
- |— _sw_src_3.json
- |— _sw_src_4.json
- |— _verified_lake_facts.md
- |— add_assistant_tags.js
- |— add_pwa_tags.js
- |— add_seo_tags.js
- |— add_theme_motion_tags.js
- |— build_assistant_kb.js
- |— build_hero_bundle.sh
- |— build_i18n_content.py
- |— build_master_en.py
- |— build_sw_lang.js
- |— fix_i18n_script_tag.py
- |— generate_dev_pdf.sh
- |— i18n_extract.py
- |— inline_material_icons.py
- |— normalize_nav.py
- |— perf_pass.js
- |— self_host_fonts.py
- |— translation_dict.py
- |— _qa_screens/
 - |— 404-desktop-footer.png
 - |— 404-desktop-mid.png
 - |— 404-desktop-top.png
 - |— 404-mobile-footer.png
 - |— 404-mobile-mid.png
 - |— 404-mobile-top.png
 - |— _findings.txt
 - |— about-desktop-footer.png
 - |— about-desktop-mid.png
 - |— about-desktop-top.png
 - |— about-mobile-footer.png
 - |— about-mobile-mid.png
 - |— about-mobile-top.png
 - |— assistant-desktop-en-countries.png
 - |— assistant-desktop-offline.png
 - |— assistant-desktop-open.png
 - |— assistant-desktop-restored.png
 - |— assistant-desktop-sw-countries.png
 - |— assistant-keyboard.png
 - |— assistant-mobile-sheet.png

- └─ careers-desktop-footer.png
- └─ careers-desktop-mid.png
- └─ careers-desktop-top.png
- └─ careers-mobile-footer.png
- └─ careers-mobile-mid.png
- └─ careers-mobile-top.png
- └─ contact-desktop-footer.png
- └─ contact-desktop-mid.png
- └─ contact-desktop-top.png
- └─ contact-mobile-footer.png
- └─ contact-mobile-mid.png
- └─ contact-mobile-top.png
- └─ fuel-desktop-footer.png
- └─ fuel-desktop-mid.png
- └─ fuel-desktop-top.png
- └─ fuel-mobile-footer.png
- └─ fuel-mobile-mid.png
- └─ fuel-mobile-top.png
- └─ gallery-desktop-footer.png
- └─ gallery-desktop-mid.png
- └─ gallery-desktop-top.png
- └─ gallery-mobile-footer.png
- └─ gallery-mobile-mid.png
- └─ gallery-mobile-top.png
- └─ globe-desktop-p0.png
- └─ globe-desktop-p0_12.png
- └─ globe-desktop-p0_3.png
- └─ globe-desktop-p0_42.png
- └─ globe-desktop-p0_55.png
- └─ globe-desktop-p0_68.png
- └─ globe-desktop-p0_8.png
- └─ globe-desktop-p0_94.png
- └─ globe-desktop-p1.png
- └─ globe-desktop-pagebottom.png
- └─ globe-desktop-pagetop.png
- └─ globe-mobile-t0.png
- └─ globe-mobile-t18.png
- └─ globe-mobile-t9.png
- └─ globe-reduced-motion.png
- └─ index-desktop-3d-finale.png
- └─ index-desktop-3d-mid.png
- └─ index-desktop-3d-start.png
- └─ index-desktop-footer.png
- └─ index-desktop-mid.png
- └─ index-desktop-top.png
- └─ index-mobile-3d-start.png
- └─ index-mobile-footer.png
- └─ index-mobile-mid.png
- └─ index-mobile-top.png
- └─ news-desktop-footer.png
- └─ news-desktop-mid.png
- └─ news-desktop-top.png
- └─ news-mobile-footer.png
- └─ news-mobile-mid.png
- └─ news-mobile-top.png
- └─ offline-mobile-top.png

```
├─ templates/  
  │├─ chat_widget.html  
  │├─ footer.html  
  │├─ mobile_nav.html  
  │└─ nav.html
```

3.1 Key Directories

Path	Contents
/*.html	29 routable pages at site root
assets/	CSS, JS, fonts, images, icons, vendor libs
assets/images/	Banners, news photos, leadership, planet textures, flags
assets/vendor/	three.module.min.js, leaflet, flexsearch (vendored, no CDN)
scripts/	Build, QA, migration, and template tooling
scripts/templates/	nav.html, mobile_nav.html, footer.html, chat_widget.html
lake-3d/	Orphaned Next.js prototype (not deployed with main site)
lake-story-assets/	Photography for our-story.html slideshow
sw.js	Service worker at root (scope = entire site)
manifest.webmanifest	PWA manifest

4. Shared Page Anatomy

4.1 Standard <head> Tags

```
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width,initial-scale=1.0" />
<meta name="description" content="..." />
<meta property="og:title" content="..." />
<meta property="og:description" content="..." />
<meta property="og:image" content="assets/images/og-cover.jpg" />
<meta property="og:type" content="website" />
<meta name="twitter:card" content="summary_large_image" />
<link rel="canonical" href="https://www.lakeoilgroup.com/page.html" />
<link rel="icon" href="favicon.ico" sizes="32x32" />
<link rel="icon" href="assets/icons/pwa/icon-192.png" type="image/png"
sizes="192x192" />
<meta name="theme-color" content="#1D3EA8" />
<link rel="manifest" href="manifest.webmanifest" />
<link rel="apple-touch-icon" href="assets/icons/pwa/apple-touch-icon.png"
/>
<link rel="preload" href="assets/fonts/files/inter-latin-400-normal.woff2"
as="font" type="font/woff2" crossorigin />
<style>... page-inline critical CSS ...</style>
<script type="application/ld+json">...</script> <!-- most pages -->
<link rel="stylesheet" href="assets/flagship.css">
<link rel="stylesheet" href="assets/assistant.css">
```

4.2 Script Load Order (Standard Page)

Order	Script	defer?	Purpose
1	assets/i18n-content.js	yes	Sets __LAKE_I18N_CONTENT__
2	assets/i18n.js	yes	Language engine
3	assets/site.js	yes	Nav, reveals, counters, tabs
4	assets/vendor/flexsearch/flexsearch.bundle.min.js	yes	Search index library
5	assets/assistant-kb.js	yes	Knowledge base payload
6	assets/assistant.js	yes	Chat assistant UI
7	[page-specific]	varies	news, leaflet, etc.
8	assets/pwa.js	yes	Service worker registration
9	assets/flagship-motion.js	yes	Motion (or motion.js on index)

Rule: i18n-content.js MUST precede i18n.js. Assistant KB MUST precede assistant.js. Motion script loads last so it can auto-tag DOM after content scripts run.

4.3 Body Structure

```

<body>
  <nav class="site-nav">...</nav>                <!-- scripts/templates/nav.html
-->
  <div class="nav-mobile" id="nav-mobile">...</div>
  <main>
    <section class="page-hero">...</section>
    <!-- page sections -->
  </main>
  <footer class="site-footer">...</footer>
  <div id="chat-widget">...</div>                <!-- assistant mount -->
  <!-- deferred scripts -->
</body>

```

4.4 Navigation Template

Primary nav (scripts/templates/nav.html) includes: Home, About, Services dropdown (fuel/lpg/lubricants/steel/concrete/logistics/containers), Network dropdown (africa map, station locator, fleet), Company dropdown (history, leadership, csr, investors, projects, gallery), News, Careers, and EN/FR/SW language switcher.

Do not hand-edit nav/footer in individual pages — run `python3 scripts/normalize_nav.py` after template changes.

4.5 data-i18n Attribute Convention

- `data-i18n="key"` — `textContent` replacement
- `data-i18n-html="key"` — innerHTML (for `<em>`, `<strong>` in translations)
- `data-i18n-placeholder`, `data-i18n-title`, `data-i18n-alt`, `data-i18n-aria`
- Keys follow dot notation: `nav.home`, `fuel.hero.title`, `chat.reply.fuel`

5. HTML Page Reference

Deep reference for every root HTML page.

404.html

Title: Page Not Found | Lake Group

Purpose: Branded not-found page with navigation back to key sections.

Approx. size: 173 lines of HTML

Sections

- .error-page — 404 message and helpful links.

CSS Loaded

flagship.css, assistant.css

JavaScript Loaded

pwa.js, flagship-motion.js, assistant stack only

Unique Features

No i18n-content.js or site.js — minimal script set. Assistant still available offline.

schema.org

Types: None

about.html

Title: About Lake Group | Our Story & Vision

Purpose: Company overview: founding story, growth narrative, values, and subsidiary introduction. Reference implementation of the Meridian (flagship) design system.

Approx. size: 585 lines of HTML

Sections

- `.page-hero` — Immersive ink band with Bebas display headline.
- `#our-story-embed` — Embedded cinematic slideshow (`our-story.html` in `iframe` or inline OSE stage).
- `.fs-split` sections — Story beats with photography from Lake Group operations.
- `.values-grid` — Mission, vision, integrity pillars.

CSS Loaded

`flagship.css`, `assistant.css`

JavaScript Loaded

Standard deferred stack + `flagship-motion.js`

Unique Features

First flagship-migrated page; template for all subsequent Meridian rollouts.

schema.org

Types: `BreadcrumbList`, `Listitem`, `Listitem`

africa-network.html

Title: Africa Network | Lake Group Operations Map

Purpose: Interactive operations map of East & Central Africa with country cards, asset legend, and Leaflet satellite map.

Approx. size: 571 lines of HTML

Sections

- `.page-hero` — Network overview headline.
- `#lake-africa-map` — Full-width Leaflet map with Esri hybrid tiles, country borders, asset markers, pipeline polylines.
- `.country-grid` — Selectable country cards syncing with map `flyTo`.
- `.map-legend` — Filter buttons by asset type: HQ, fuel, port, container, industrial, logistics.

CSS Loaded

`flagship.css`, `assistant.css`, `vendor/leaflet/leaflet.css`

JavaScript Loaded

`leaflet.js`, `data_countries_africa.js`, `africa-network-map.js` + standard stack

Unique Features

Loads Leaflet synchronously (no defer) plus `data_countries_africa.js` before `africa-network-map.js`. Requires network for tile layers.

schema.org

Types: `BreadcrumbList`, `ListItem`, `ListItem`

careers.html

Title: Careers at Lake Group | Jobs in East Africa

Purpose: Culture, benefits, CV submission form (mock).

Approx. size: 288 lines of HTML

Sections

- Build Your Career Across Africa — Primary content block.
- Drop Us Your CV — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

concrete.html

Title: Concrete & Aggregate | GCCP | Lake Group

Purpose: GCCP ready-mix concrete and aggregate — Lugoba quarry, project portfolio.

Approx. size: 311 lines of HTML

Sections

- Building East Africa's Future — Primary content block.
- Our Plant in Action — Primary content block.
- Specialist Concrete Portfolio — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: Service, Organization, BreadcrumbList, ListItem, ListItem

schema.org

Types: Service, Organization, BreadcrumbList, ListItem, ListItem

contact.html

Title: Contact Lake Group | Get in Touch

Purpose: HQ address, phone, email, regional offices, contact form.

Approx. size: 287 lines of HTML

Sections

- We're Happy to Hear From You — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: ContactPage, Organization, ContactPoint,
BreadcrumbList, ListItem, ListItem

schema.org

Types: ContactPage, Organization, ContactPoint, BreadcrumbList, ListItem,
ListItem

container-services.html

Title: Container Services | AFICD & ACFS | Lake Group

Purpose: AFICD inland container depots — Tanzania, Zambia, Mozambique.

Approx. size: 277 lines of HTML

Sections

- The Inland Gateway to East African Trade — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: Service, Organization, BreadcrumbList, ListItem, ListItem

schema.org

Types: Service, Organization, BreadcrumbList, ListItem, ListItem

csr.html

Title: CSR & Sustainability | Lake Group

Purpose: Corporate social responsibility programs and community investment.

Approx. size: 277 lines of HTML

Sections

- Responsible Growth. Lasting Impact. — Primary content block.
- How We Give Back — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

dashboard.html

Title: Client Dashboard | Lake Group

Purpose: Mock client portal UI (sign-in form, dashboard preview). Demo only — excluded from sitemap and robots.

Approx. size: 337 lines of HTML

Sections

- .auth-panel — Email/password mock sign-in.
- .dashboard-preview — Placeholder shipment tracking widgets.

CSS Loaded

flagship.css, assistant.css

JavaScript Loaded

Standard stack

Unique Features

robots.txt Disallow. No JSON-LD. Not in assistant KB page map.

schema.org

Types: None

fleet.html

Title: Our Fleet | Lake Trans | Lake Group

Purpose: 700+ truck fleet specifications and capabilities.

Approx. size: 291 lines of HTML

Sections

- Purpose-Built for Every Load — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

fuel.html

Title: Fuel & Petroleum | Lake Group

Purpose: Lake Oil petroleum distribution — 85+ stations, bulk supply, competitive advantages.

Approx. size: 284 lines of HTML

Sections

- Keeping Africa Moving — Primary content block.
- Our Competitive Edge — Primary content block.

CSS Loaded

`assets/flagship.css, assets/assistant.css`

JavaScript Loaded

`assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js`

Unique Features

Migrated Meridian design. JSON-LD: Service, Organization, BreadcrumbList, ListItem, ListItem

schema.org

Types: Service, Organization, BreadcrumbList, ListItem, ListItem

gallery.html

Title: Gallery | Lake Group

Purpose: Masonry photo gallery with lightbox from operations and events.

Approx. size: 442 lines of HTML

Sections

- Lake Group in Action — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

history.html

Title: Our History | Lake Group Since 2006

Purpose: Timeline from 2006 founding to present pan-African conglomerate.

Approx. size: 285 lines of HTML

Sections

- Two Decades of Growth — Primary content block.
- Be Part of the Next Chapter — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

index.html

Title: Lake Group | Powering East & Central Africa

Purpose: Corporate homepage and primary entry point. Presents the Lake Group value proposition, live 3D global operations experience, division grid, founder message, stats, news teaser, and primary CTAs.

Approx. size: 2931 lines of HTML

Sections

- `#fuel-experience / .experience-3d` — Full-viewport interactive Three.js Earth globe with autonomous 39-second loop showing verified operational footprint across 8 countries + Dubai.
- `.hero` — Above-the-fold headline, subcopy, and primary navigation into services and contact.
- `.stats-grid` — Animated counters: 4,600+ employees, 700+ trucks, 85+ stations, 8 countries.
- `.divisions-grid` — Eight sector cards linking to fuel, LPG, lubricants, steel, concrete, logistics, containers, and services hub.
- `.founder-section` — Pull quote and portrait of founder Ally Edha Awadh.
- `.cta-band` — Final conversion band with contact and careers links.

CSS Loaded

`assets/theme.css, assets/assistant.css`

JavaScript Loaded

`i18n stack, site.js, assistant stack, lazy hero-3d.bundle.js, pwa.js, motion.js`

Unique Features

Only page using `theme.css` + `motion.js` (not flagship). Lazy-loads `hero-3d.bundle.js` via `IntersectionObserver` with 600px `rootMargin`. Inline critical CSS in `<head>` (~2000 lines). Organization JSON-LD.

schema.org

Types: `Organization`, `Person`, `PostalAddress`

investors.html

Title: Investor Relations | Lake Group

Purpose: Financial highlights, governance, currency-convertible figures.

Approx. size: 269 lines of HTML

Sections

- A Track Record of Vertical Growth — Primary content block.
- What Drives Our Value — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

leadership.html

Title: Leadership | Lake Group

Purpose: Executive team, founder profile, subsidiary management.

Approx. size: 291 lines of HTML

Sections

- Guided by Experience. Driven by Vision. — Primary content block.
- Ally Edha Awadh — Primary content block.
- Group & Lake Oil Management — Primary content block.
- Companies Under Lake Group — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

[schema.org](#)

Types: BreadcrumbList, ListItem, ListItem

logistics.html

Title: Transport & Haulage | Lake Trans | Lake Group

Purpose: Lake Trans bulk liquid and dry cargo haulage.

Approx. size: 301 lines of HTML

Sections

- The Backbone of East Africa's Supply Chain — Primary content block.
- Purpose-Built for Africa's Roads — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: Service, Organization, BreadcrumbList, ListItem, ListItem

schema.org

Types: Service, Organization, BreadcrumbList, ListItem, ListItem

lpg.html

Title: LPG Gas | Lake Gas | Lake Group

Purpose: Lake Gas LPG — cylinder sizes, bottling plants, clean cooking access.

Approx. size: 290 lines of HTML

Sections

- Clean Energy for Every Kitchen — Primary content block.
- Lake Gas Across Africa — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: Service, Organization, BreadcrumbList, ListItem, ListItem

schema.org

Types: Service, Organization, BreadcrumbList, ListItem, ListItem

lubricants.html

Title: Lubricants | Lake Lubes | Lake Group

Purpose: Lake Lubes — greases, industrial and automotive lubricants.

Approx. size: 267 lines of HTML

Sections

- Engineered for Performance — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: Service, Organization, BreadcrumbList, ListItem, ListItem

schema.org

Types: Service, Organization, BreadcrumbList, ListItem, ListItem

media-center.html

Title: Media Center | Lake Group

Purpose: Press releases and media kit downloads.

Approx. size: 272 lines of HTML

Sections

- Latest Announcements — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

news-article.html

Title: News Article | Lake Group

Purpose: Single article view driven by ?id= query parameter.

Approx. size: 243 lines of HTML

Sections

- #article-root — Title, date, category badge, banner, description paragraphs, image gallery, optional YouTube embed.

CSS Loaded

flagship.css, assistant.css

JavaScript Loaded

Standard + news-data.js + news.js

Unique Features

NewsArticle JSON-LD generated dynamically. Redirects to news.html if invalid id.

schema.org

Types: BreadcrumbList, ListItem, ListItem, ListItem

news.html

Title: News & Events | Lake Group

Purpose: News listing with search, category filter, country filter, and paginated article cards.

Approx. size: 300 lines of HTML

Sections

- `.page-hero` — News & Events headline.
- `.news-filters` — Search input, category select, country select.
- `#news-list` — Dynamically rendered by `news.js` from `LAKE_NEWS` array.

CSS Loaded

`flagship.css`, `assistant.css`

JavaScript Loaded

Standard + `news-data.js` + `news.js`

Unique Features

Loads `news-data.js` + `news.js`. Article cards use `data-category` and `data-countries` for client-side filtering.

schema.org

Types: `CollectionPage`, `BreadcrumbList`, `ListItem`, `ListItem`

offline.html

Title: You're Offline | Lake Group

Purpose: PWA offline fallback served by service worker when navigation fails without network.

Approx. size: 215 lines of HTML

Sections

- `.offline-shell` — Explains offline state; links to precached about, services, contact, africa-network.

CSS Loaded

`flagship.css`, `assistant.css`

JavaScript Loaded

`pwa.js`, `flagship-motion.js`, `assistant stack`

Unique Features

Listed in `robots.txt` Disallow. Precached in `sw.js` `PRECACHE_URLS`.

schema.org

Types: None

our-story.html

Title: Lake Group: Our Story

Purpose: Standalone cinematic brand story — eight auto-advancing scenes with Lake Group photography, keyboard/tap navigation.

Approx. size: 619 lines of HTML

Sections

- .ose-stage — Full-screen slideshow scenes 1–8.
- .story-stats — Closing stats overlay with link back to main site.

CSS Loaded

flagship.css, assistant.css

JavaScript Loaded

i18n (sync), pwa.js, flagship-motion.js, assistant

Unique Features

Not linked from main nav. i18n scripts loaded without defer. Uses lake-story-assets/ images.

schema.org

Types: None

projects.html

Title: Major Projects | Lake Group

Purpose: Major infrastructure and industrial project case studies.

Approx. size: 249 lines of HTML

Sections

- Delivering East Africa's Development — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

services.html

Title: Services | Lake Group

Purpose: Hub page linking all eight business sectors.

Approx. size: 375 lines of HTML

Sections

- One Integrated Vision — Primary content block.
- Ready to Work Together? — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

station-locator.html

Title: Station Locator | Lake Oil | Lake Group

Purpose: Lake Oil retail station finder across Tanzania.

Approx. size: 310 lines of HTML

Sections

- 85+ Stations Across Tanzania — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

steel.html

Title: Lake Steel | HS-CR Rebars | Lake Group

Purpose: Lake Steel HS-CR rebars — 100,000 MT/year rolling mill, first in Tanzania.

Approx. size: 295 lines of HTML

Sections

- First in Tanzania. Built to Last. — Primary content block.
- From Billets to Bars — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: Service, Organization, BreadcrumbList, ListItem, ListItem

schema.org

Types: Service, Organization, BreadcrumbList, ListItem, ListItem

sustainability.html

Title: Sustainability | Lake Group

Purpose: Environmental stewardship, LPG clean cooking, ESG metrics.

Approx. size: 251 lines of HTML

Sections

- Energy for Africa. Responsibly. — Primary content block.
- Sustainability by the Numbers — Primary content block.

CSS Loaded

assets/flagship.css, assets/assistant.css

JavaScript Loaded

assets/i18n-content.js, assets/i18n.js, assets/site.js,
assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js,
assets/assistant.js, assets/pwa.js, assets/flagship-motion.js

Unique Features

Migrated Meridian design. JSON-LD: BreadcrumbList, ListItem, ListItem

schema.org

Types: BreadcrumbList, ListItem, ListItem

6. JavaScript Modules

assets/site.js

Shared site behaviour module (IIFE). Responsibilities: • `initNav()` — mobile toggle, exact-filename active link highlighting • `initReveal()` — `IntersectionObserver` for `.reveal` elements • `initCounters()` — animated `[data-count]` with easing • `initTabs()` — `.tab-nav` / `.tab-pane` switching • `initChat()` — legacy canned-reply chat (bypassed when assistant active) • `initAnchors()` — smooth scroll for hash links • `initForms()` — mock form submit UX for data-mock forms • `initCurrency()` — investors page currency converter Exports `window.LakeSite` with `initReveal/initCounters`. Calls `LakeI18n.init()` on `DOMContentLoaded`.

assets/i18n.js

IIFE exposing `window.LakeI18n`. Languages: `en`, `fr`, `sw`. Reads `window.__LAKE_I18N_CONTENT__` from `i18n-content.js` (never fetch — `file://` safe). Attributes: `data-i18n`, `data-i18n-html`, `data-i18n-placeholder`, `data-i18n-title`, `data-i18n-alt`, `data-i18n-aria`. Persists `lake-lang` in `localStorage`. Dispatches `lake-i18n-applied` event.

assets/i18n-content.js

Build artifact: `window.__LAKE_I18N_CONTENT__ = { en: {...}, fr: {...}, sw: {...} }`. ~1,442 keys. Generated by `scripts/build_i18n_content.py` from `scripts/_master_en.json` + `translation_dict.py`.

assets/pwa.js

Registers sw.js relative to script URL. Shows branded update toast when new SW version waiting. SKIP_WAITING on user click. Reload only after user-initiated update.

assets/motion.js

Pairs with theme.css. Adds html.lg-motion. Auto-tags .reveal on grids/orphans. Nav hide/show on scroll. Card 3D tilt (pointer:fine). Respects prefers-reduced-motion.

assets/flagship-motion.js

Pairs with flagship.css. Adds html.fs-motion. Richer fx-* choreography (fx-rise, fx-clip, fx-mask, fx-wipe). Scroll progress via [data-fx-scroll]. Magnetic CTA hover. [data-fs-count] counters.

assets/hero-3d.js

Three.js module (bundled to hero-3d.bundle.js). See Section 9.

assets/hero-3d.bundle.js

esbuild output: hero-3d.js + three.module.min.js as classic script. ~0.5MB. Rebuild via scripts/build_hero_bundle.sh.

assets/assistant.js

Offline knowledge assistant. FlexSearch retrieval over assistant-kb.js. IndexedDB lake-assistant/messages. Sets `__LAKE_ASSISTANT_ACTIVE__`. UI replaces `#chat-widget`.

assets/assistant-kb.js

Build artifact: `window.__LAKE_ASSISTANT_KB__`. Generated by `scripts/build_assistant_kb.js`.

assets/news.js

Renders news list and article detail from `window.LAKE_NEWS`. Filters by `search/category/country`. YouTube embed helper.

assets/news-data.js

`window.LAKE_NEWS` array — article metadata, images, categories, dates.

assets/africa-network-map.js

Leaflet map init. `ASSETS`, `PIPELINES`, `COUNTRY_META`. `window.LakeAfricaMap` API. GeoJSON from `__LAKE_AFRICA_GEOJSON__`.

assets/data_countries_africa.js

`window.__LAKE_AFRICA_GEOJSON__` — Africa country borders for offline/file:// map.

6.1 Vendor Libraries

File	Version	License	Usage
vendor/three.module.min.js	r16x	MIT	hero-3d.js WebGL rendering
vendor/leaflet/leaflet.js	1.x	BSD-2	africa-network.html map
vendor/flexsearch/flexsearch.bundle.min.js	0.7.x	Apache-2.0	assistant.js full-text search

All vendor code is self-hosted — no runtime CDN dependencies except map tile servers (Esri, OSM, OpenTopoMap) on africa-network.html.

7. CSS Design Systems

7.1 theme.css vs flagship.css

Aspect	theme.css (Legacy)	flagship.css (Meridian)
Used on	index.html only	All other migrated pages (28)
Design language	Blue/yellow corporate, 4–10px radius	Ink/paper editorial, 0–2px radius, Bebas display
Dark sections	.section-dark (navy)	.fs-on-dark / --ink system
Signature motif	Yellow divider bars	Meridian tick + corner brackets
Motion partner	motion.js (.reveal)	flagship-motion.js (.fx-*)
Typography	Inter + system	Bebas Neue display + Inter body + Playfair quotes
Tokens	--blue, --yellow, --radius 4px	--ink, --paper, --gold, --sp-* scale

7.2 motion.js vs flagship-motion.js

Feature	motion.js	flagship-motion.js
Gate class	html.lg-motion	html.fs-motion
Reveal classes	.reveal, .reveal-left, .reveal-scale	.fx, .fx-rise, .fx-clip, .fx-mask, .fx-wipe
Auto-tagger	Grid children + orphans	Richer variant selection by element type
Nav behaviour	Hide on scroll down	Same + elevation shadow
Card tilt	3D pointer tilt (~3°)	Not primary — magnetic CTA instead
Scroll effects	—	[data-fx-scroll] → --fxp property
Counters	site.js [data-count]	[data-fs-count] with expo-out easing
Skip zones	#fuel-experience, .experience-3d, nav, chat	Same + .page-hero

7.3 assistant.css

Isolated stylesheet for the knowledge assistant panel. Loaded on every page alongside the main design system. Uses CSS variables from whichever theme is active.

7.4 Migration Rule

A migrated page loads **either** theme.css+motion.js **or** flagship.css+flagship-motion.js — never both. See FLAGSHIP_DESIGN.md for component inventory and migration checklist.

8. 3D Hero Architecture (hero-3d.js)

Interactive WebGL globe on index.html — verified operational footprint only (scripts/_verified_lake_facts.md).

8.1 File Pipeline

1. assets/hero-3d.js — ES module source with `import * as THREE`
2. scripts/build_hero_bundle.sh — esbuild bundles module + three into classic script
3. assets/hero-3d.bundle.js — production payload (~0.5MB)
4. index.html lazy-loads bundle when #fuel-experience enters viewport (+600px margin)

8.2 Four Chapters (39s Loop)

Chapter	Progress	Duration	Content
1 Planet	0.00 – 0.30	~10s	Earth at distance; Africa rotates into view
2 Footprint	0.30 – 0.65	~12s	9 sites ignite from Dar HQ; route arcs draw outward
3 Operations	0.65 – 0.90	~9s	Camera over East Africa; facility callouts on leader lines
4 Identity	0.90 – 1.00	~3s + 5s hold	Pull-back; DOM finale overlay (logo + tagline); fade to black; restart

8.3 Verified Sites (SITES array)

Key	City	Country	Notes
tz	Dar es Salaam · HQ	Tanzania	Plot 49 Mikocheni — hub site
ke	Nairobi	Kenya	Kenya network base
rw	Kigali	Rwanda	Lake Petroleum Rwanda
bu	Bujumbura	Burundi	Burundi Petroleum
cd	Lubumbashi	DR Congo	DRC Petroleum
zm	Ndola	Zambia	Lake Petroleum Zambia
et	Addis Ababa	Ethiopia	Wadi Elsundus / WAS
mz	Beira	Mozambique	Lake Oil LDA + AFICD
ae	MERM · SAFF	Dubai · UAE	Middle East operations

8.4 Facility Callouts (Chapter 3)

- **TANGA LPG TERMINAL** — 3,000 MT storage, marine mooring buoy
- **DAR ES SALAAM PORT** — Vessel bunkering, 38M-litre depot
- **LAKE STEEL · KIBAHA** — HS-CR mill, 100,000 MT/yr
- **GCCP · DAR ES SALAAM** — Ready-mix, Lugoba quarry

8.5 Textures (assets/images/planet/)

File	Resolution	Usage
earth_color_2048.jpg	2048	Diffuse map (NASA Blue Marble)
earth_normal_1024.jpg	1024	Normal map (skipped on low quality)
earth_specular_1024.jpg	1024	Ocean specular mask
earth_clouds_1024.jpg	1024	Drifting cloud alpha layer

TEX_ROT_Y = -120° aligns equirectangular Greenwich with three.js sphere UV mapping.

8.6 Performance Optimizations

- **Quality tiers:** isLow = width<768 OR hardwareConcurrency<4 → fewer sphere segments, no normal map, lower star count, pixel ratio capped at 1

- **maybeDemoteQuality()**: Sustained >33ms frames for 90 frames → pixel ratio 1, clouds hidden
- **IntersectionObserver**: Pauses rAF when #experience-3d-panel off-screen
- **visibilitychange**: Pauses when tab hidden
- **InstancedMesh**: 9 country markers as single draw call
- **Canvas sprite labels**: Generated once at init, not per-frame
- **Lazy bundle load**: Does not block initial page load
- **prefers-reduced-motion**: Shows static finale overlay, no WebGL
- **SW routing**: network-first for hero-3d.bundle.js (never stale cache)

8.7 Public Hooks

`window.__lake3dFrames`, `window.__lake3dProgress` — used by QA scripts (`_globe_qa2.js`).

9. Progressive Web App

9.1 manifest.webmanifest

Field	Value
name / short_name	Lake Group
start_url	./index.html
scope	./
display	standalone
theme_color	#1D3EA8
background_color	#0E1F5A
icons	192, 512, maskable 192/512 in assets/icons/pwa/

9.2 sw.js Versioning

VERSION constant (currently v10) — bump on every deploy that changes precached files.
Activate handler deletes all lake- * caches not in KNOWN_CACHES.

9.3 Cache Buckets

Cache Name	Purpose
lake-precache-{VERSION}	Install-time precache (shell, fonts, core JS/CSS)
lake-pages-{VERSION}	HTML navigations (network-first)
lake-images-{VERSION}	Images (cache-first, max 150 entries LRU)
lake-assets-{VERSION}	JS/CSS/fonts/vendor (stale-while-revalidate)

9.4 Fetch Routing Strategies

Route Class	Strategy	Examples
navigate	Network-first → cached page → offline.html	All HTML pages
network-first-asset	Network-first	news-data.js, hero-3d.bundle.js
cache-first-asset	Cache-first	/assets/fonts/, /assets/vendor/
cache-first-image	Cache-first + LRU trim	Most /assets/images/
swr-image	Stale-while-revalidate	n-slider, lake-story-assets, products/
swr-asset	Stale-while-revalidate	site.js, i18n, assistant, manifest

9.5 pwa.js Behaviour

- Registers SW on window load
- Resolves sw.js URL relative to pwa.js (subpath-safe)
- Update toast: navy background, gold border, "Refresh" triggers SKIP_WAITING
- controllerchange reload only if user clicked Refresh (no surprise reloads)
- Silently no-ops on file:// or unsupported browsers

10. Internationalization (i18n)

10.1 Supported Languages

English (en), French (fr), Swahili (sw). Portuguese was removed; SW replaced PT in the language switcher.

10.2 Pipeline

```
HTML pages (data-i18n attributes)
  |
  ▼
scripts/i18n_extract.py → discovers keys / orphans
  |
  ▼
scripts/build_master_en.py → scripts/_master_en.json (English master)
  |
  ▼
scripts/translation_dict.py → PHRASES_FR, PHRASES_SW, TERMS_*
  |
  ▼
scripts/build_i18n_content.py → assets/i18n-content.json
                                → assets/i18n-content.js (runtime)
  |
  ▼
assets/i18n.js → applies to DOM on load + language switch
```

10.3 Rebuild Commands

```
python3 scripts/build_master_en.py
python3 scripts/build_i18n_content.py
```

10.4 Coverage Notes

- ~1,442 translation keys across the site
- Full coverage: nav, footer, chat/assistant, homepage hero
- Partial coverage: long-tail page content falls back to English DOM text gracefully
- hero-3d.js reads LakeI18n.current for SITE_NAMES country labels

10.5 QA

`scripts/_qa_check_i18n.js` — validates key coverage. `scripts/build_sw_lang.js` — language-specific SW precache variants.

11. Knowledge Assistant System

11.1 Architecture

```
scripts/build_assistant_kb.js
├─ reads assets/i18n-content.json (page chunks)
├─ merges CURATED_FACTS from _verified_lake_facts.md
└─ writes assets/assistant-kb.js

assets/assistant.js
├─ FlexSearch index per language (en/fr/sw)
├─ IndexedDB persistence (lake-assistant/messages)
├─ Retrieves verbatim passages – never generates text
└─ "Read more →" links to source page
```

11.2 Document Shape

Each KB document: { id, t, s, u, k, f } — title, text, url, keywords, fact-boost flag.

11.3 Chunking

MIN_LEN=30 chars, CHUNK_TARGET=340 chars. Keys grouped by page prefix (fuel.*, about.*, etc.).

11.4 Rebuild

```
node scripts/build_assistant_kb.js
```

Run after i18n content changes or curated fact updates.

11.5 Interaction with site.js

window.__LAKE_ASSISTANT_ACTIVE__ = true set at assistant.js parse time prevents legacy initChat() from binding duplicate handlers.

12. News System

12.1 Data Model (news-data.js)

window.LAKE_NEWS — array of articles:

```
{
  id: number,
  title: string,
  date: string,
  category: string,      // e.g. "Expansion", "CSR", "Events"
  bannerImage: string,
  description: string[], // paragraphs
  images: string[],      // gallery paths
  video: string|null     // YouTube URL
}
```

12.2 news.html

- Renders #news-list via renderNewsList()
- Filters: #news-search (text), #news-category, #news-country
- Country detection via keyword heuristics in articleCountries()
- Re-inits reveal animations after filter

12.3 news-article.html

- Reads ?id= or ?newsid= query param
- Renders full article with image grid and optional YouTube iframe
- Updates document title and meta description dynamically
- Injects NewsArticle JSON-LD

12.4 Images

Stored under assets/images/news/{id}/photo_*.jpg. SW uses network-first for news-data.js.

13. Africa Network Map (Leaflet)

13.1 Dependencies

Loaded synchronously (before deferred stack) on africa-network.html:

1. leaflet.css + leaflet.js
2. data_countries_africa.js → window.__LAKE_AFRICA_GEOJSON__
3. africa-network-map.js

13.2 Tile Layers

- **Hybrid (default):** Esri World Imagery + Transportation + Boundaries overlays
- Satellite Imagery, Terrain (OpenTopoMap), Streets (OSM)

13.3 Data Layers

- **countryLayer:** GeoJSON fills for 8 ops countries + neighbours
- **assetLayer:** 22 circle markers by type (hq, fuel, port, container, industrial, logistics)
- **pipelineLayer:** TAZAMA pipeline, Northern corridor, Southern route polylines

13.4 Public API

```
window.LakeAfricaMap = {
  flyToCountry(id, cardEl),
  filterAssets(type), // 'all' | 'hq' | 'fuel' | ...
  resetView()
};
window.selectCountry(id, card) // called from country cards
```

13.5 Offline / file://

GeoJSON via script tag (not fetch). Tiles require network. Fallback: coloured circles if GeoJSON missing.

14. Build & Maintenance Scripts

Script	Purpose
build_i18n_content.py	Generate i18n-content.json + .js from master + translations
build_master_en.py	Extract English strings into _master_en.json
translation_dict.py	FR/SW phrase and term dictionaries
i18n_extract.py	Scan HTML for data-i18n keys and orphans
normalize_nav.py	Inject templates into all HTML pages
build_assistant_kb.js	Generate assistant-kb.js knowledge base
build_hero_bundle.sh	esbuild hero-3d.js → hero-3d.bundle.js
build_sw_lang.js	Language-variant service worker assets
add_pwa_tags.js	Inject manifest/theme-color links
add_seo_tags.js	Inject OG, canonical, description tags
add_theme_motion_tags.js	Inject CSS/JS link tags
add_assistant_tags.js	Inject assistant script/css tags
self_host_fonts.py	Download and wire self-hosted font files
inline_material_icons.py	Inline Material Symbols SVGs
perf_pass.js	Performance audit pass on pages
_qa_check_links.js	Broken link checker
_qa_check_i18n.js	i18n coverage QA
_qa_matrix.js	Cross-page QA matrix
_qa_overflow_check.js	Horizontal overflow detection
_globe_qa.js / _globe_qa2.js	3D hero automated QA
_assistant_qa.js	Assistant retrieval QA
generate_dev_pdf.sh	PDF generation helper for docs

15. SEO & Structured Data

15.1 robots.txt

```
User-agent: *
Allow: /
Disallow: /offline.html
Disallow: /dashboard.html

Sitemap: https://www.lakeoilgroup.com/sitemap.xml
```

15.2 sitemap.xml

Lists all public pages at lakeoilgroup.com. Priority 1.0 for homepage, 0.8 for news, 0.7 for content pages. lastmod: 2026-07-04.

Excluded: offline.html, dashboard.html, 404.html, our-story.html (not in sitemap).

15.3 JSON-LD Types by Page

@type	Pages
Organization	index.html
WebPage	Most content pages
NewsArticle	news-article.html (dynamic)
BreadcrumbList	Service/company pages

15.4 Open Graph

og:title, og:description, og:image (assets/images/og-cover.jpg), og:type=website on all major pages. twitter:card=summary_large_image.

15.5 Canonical URLs

Each page includes <link rel="canonical" href="https://www.lakeoilgroup.com/{page}.html"> — added/maintained by add_seo_tags.js.

16. Deployment Checklist

16.1 Pre-Deploy

1. Run i18n pipeline if copy changed: `build_master_en.py` → `build_i18n_content.py`
2. Run `node scripts/build_assistant_kb.js` if content/facts changed
3. Run `bash scripts/build_hero_bundle.sh` if hero-3d.js changed
4. Run `python3 scripts/normalize_nav.py` if templates changed
5. Bump `VERSION` in `sw.js` if any precached asset changed
6. Bump `?v=N` on `hero-3d.bundle.js` script in `index.html` if bundle changed
7. Update `sitemap.xml` lastmod dates for changed pages
8. Run QA: `node scripts/_qa_check_links.js, _qa_check_i18n.js`

16.2 Local Testing

```
npm install          # firebase-tools devDependency
npm run serve        # Firebase hosting emulator
# OR any static server:
python3 -m http.server 8080
```

16.3 Deploy

```
npm run deploy      # firebase deploy --only hosting
```

Requires Firebase project configuration (`firebase.json` + `.firebaserc` — configure locally, not committed).

16.4 Post-Deploy Verification

- Hard-refresh homepage — verify 3D hero loads after scroll
- Switch EN/FR/SW — verify nav and hero translate
- Open assistant — query "fuel stations Tanzania"
- DevTools → Application → Service Workers — confirm new `VERSION` active
- Test offline: load site, go offline, navigate to `about.html`
- Validate structured data: Google Rich Results Test

- Check [africa-network.html](#) map tiles load

17. Accessibility

17.1 Reduced Motion

- `prefers-reduced-motion: reduce` — `motion.js/flagship-motion.js` skip `html.lg-motion/fs-motion`; all `.reveal/.fx` visible immediately
- `hero-3d.js` — static finale overlay instead of WebGL animation
- CSS transitions disabled under reduced motion media query in both design systems

17.2 Semantic HTML

- `<nav>`, `<main>`, `<footer>`, `<article>` landmarks
- Heading hierarchy `h1` → `h2` → `h3` per page
- Alt text on images; `data-i18n-alt` for translated alts

17.3 Keyboard & ARIA

- Nav toggle: `aria-label="Menu"`
- Language switcher: `aria-label="Language"`
- Assistant: `aria-label` on open/close, `role="status"` on PWA toast
- Chat input: `data-i18n-aria` for translated `aria-label`
- `our-story.html`: `Space/→` keyboard navigation

17.4 Colour Contrast

Flagship system uses `--gold-deep (#C79B00)` for WCAG-safe gold on light surfaces. Ink bands use `--ink-text` at 78% opacity minimum.

17.5 Focus Management

Buttons and links have visible focus styles in `flagship.css`. Assistant panel traps focus when open on mobile sheet layout.

18. Appendices

18.1 Brand Colour Reference

Name	Hex	Usage
Navy	#0E1F5A	Headers, text, PWA toast, ink-adjacent
Blue	#1D3EA8	Primary brand, theme-color
Gold	#FFD700	Accents, CTAs, meridian ticks
Red	#CC1E1E	Fuel markers, alerts
Ink	#070C1E	Flagship dark bands
Paper	#F6F5F1	Flagship light sections

18.2 Contact & HQ (Verified)

Plot 49, Mikocheni Light Industrial Area, Dar es Salaam, Tanzania
Tel: +255 222 780 510 · Email: admin@lakeoilgroup.com

18.3 Related Documentation

- README.md — file:// fixes, lake-3d status, i18n overview
- FLAGSHIP_DESIGN.md — Meridian migration guide
- scripts/_verified_lake_facts.md — source of truth for 3D sites and assistant facts
- QA_REPORT.md — latest QA findings

18.4 Glossary

MPA

Multi-Page Application — each URL is a separate HTML file

Meridian

Flagship design system (flagship.css + flagship-motion.js)

OSE

Our Story Experience — cinematic slideshow in our-story.html

SWR

Stale-While-Revalidate — cache strategy serving cached then updating

KB

Knowledge Base — assistant-kb.js retrieval documents

8A. hero-3d.js — Implementation Deep Dive

8A.1 Entry Point and Boot Sequence

The bundled script exposes no exports. On load it calls `boot()` which:

1. Locates `#experience-3d-panel` inside `#fuel-experience`
2. Checks `prefers-reduced-motion` — if set, adds `show-finale show-finale-full` classes and returns
3. Calls `initHero3D(mount)` inside `try/catch`; on failure shows `.experience-3d-error`
4. Sets `mount.classList.add('is-ready')` when WebGL context is live

8A.2 Coordinate System

The globe uses a unit sphere of radius `GLOBE_R = 3.4`. Lat/lon conversion via `latLonTo(lat, lon, r, out)`:

```
phi = degToRad(90 - lat)      // colatitude from north pole
theta = degToRad(lon) - GLOBE_CENTER_LON  // GLOBE_CENTER_LON = 30°E
x = r * sin(phi) * sin(theta)
y = r * cos(phi)
z = r * sin(phi) * cos(theta)
```

All markers, arcs, and callout leader lines are positioned in this space then parented to the `globe` Group so they rotate with the planet during Chapter 1.

8A.3 Renderer Configuration

Setting	High Quality	Low Quality (isLow)
antialias	true	false
pixelRatio	min(DPR, 1.5)	min(DPR, 1)
Earth segments	72×48	48×32
Normal map	loaded	skipped
Star count	650 + 160	350 + 80
toneMapping	ACESFilmicToneMapping	same
toneMappingExposure	1.15 (modulated by loop fade)	same

8A.4 Route Arcs

Great-circle arcs connect Dar es Salaam HQ (index 0) to each remote site. Each arc is a `THREE.Mesh` with `TubeGeometry` along a quadratic Bezier curve elevated above the surface. Arc visibility uses `setDrawRange()` to animate drawing — only the portion corresponding to chapter progress renders, creating the "ignite outward" effect in Chapter 2.

Arc timing is staggered: Dubai (ae) lands last as the most distant connection. Hub ring at HQ pulses with sinusoidal scale during hub ignition phase (`HUB_ON_START` to `HUB_ON_END`).

8A.5 Label Sprites

Each of 9 sites gets a `THREE.Sprite` with a canvas-generated texture:

- Supersampled at 3× for crisp text on retina displays
- Dark plate with gold stroke for HQ, blue-grey for subsidiaries
- `anchor` property per site fans labels in screen space to prevent overlap in dense East Africa cluster
- Opacity driven by chapter progress; dimmed during Chapter 3 except hovered marker and HQ
- Re-localized country names via `SITE_NAMES[lang]` when `lake-i18n`-applied fires

8A.6 Facility Callouts (Leader Lines)

Four verified facilities use a three-part visual: surface anchor dot, quadratic leader line in 3D, and title sprite offset by `off: [east, north, up]` in surface-tangent space. Each callout fades in during Chapter 3 (`fadeStart` staggered) and fades out before loop reset.

8A.7 Camera Rig

CAM_KEYS array defines (azimuth, elevation, distance×R, fov) at progress keyframes.

sampleCamera(gp, out) finds the surrounding keyframe pair and smoothsteps between them.

Mouse parallax adds $\pm 6\%$ of camera distance on pointer move (desktop only).

8A.8 Loop Timing Mathematics

```
LOOP_DURATION = 34 seconds // active animation
LOOP_PAUSE = 5 seconds // hold on finale
Total cycle = 39 seconds

gp = min(1, cycleT / LOOP_DURATION) // progress 0..1

Chapter 1: gp in [0, 0.30]
Chapter 2: gp in [0.30, 0.65]
Chapter 3: gp in [0.65, 0.90]
Chapter 4: gp in [0.90, 1.00]

Fade at seam: getLoopFade() combines fadeOut(last 0.9s) and fadeIn(first 0.8s)
```

8A.9 Memory Management

initHero3D returns a teardown function that: cancels rAF, removes event listeners, disconnects ResizeObserver and IntersectionObserver, traverses scene disposing geometries/materials/textures, and calls `renderer.dispose()`. The bundle does not currently invoke teardown on SPA navigation (not applicable — full page loads).

8A.10 Editing Workflow

1. Edit assets/hero-3d.js (ES module with import)
2. Run `bash scripts/build_hero_bundle.sh` (requires esbuild)
3. Bump `?v=N` query on index.html script injector
4. Bump `VERSION` in sw.js (network-first but version bump clears confusion)
5. Run `node scripts/_globe_qa2.js` for automated frame capture QA

7A. Meridian Component Catalog

Complete reference for flagship.css components. Never apply to #fuel-experience or .experience-3d-*.

7A.1 Typography Classes

Class	Font	Usage
.fs-hero-title	Bebas Neue	Page hero headlines, clamp 4–9.5rem
.fs-display	Bebas Neue	Section display headings
.fs-eyebrow	Inter 600	Tracked uppercase label with meridian tick
.fs-lede	Inter	Intro paragraph, larger size
.fs-serif-quote	Playfair italic	Pull quotes
.fs-outline / .fs-outline-gold	Bebas	Stroke-outlined display text
.section-title	Bebas (restyled)	Legacy class, mapped to flagship

7A.2 Layout Primitives

Class	Behaviour
.fs-section / .section	Vertical padding using --sp-* scale (96px default)
.fs-on-dark / .section-dark	Ink background band with light text tokens
.fs-split	5fr/7fr two-column, reverses at 880px
.fs-split-even	1fr/1fr equal split
.grid-2/3/4	Responsive grids, collapse at 720px
.fs-stat-rail	Horizontal KPI strip with meridian baselines

7A.3 Interactive Components

Class	States
.btn-primary	Gold fill, notched corner, magnetic hover via flagship-motion.js
.btn-blue	Brand blue fill
.btn-outline / .btn-ghost	For use on dark (.fs-on-dark) backgrounds
.btn-outline-dark	For use on light backgrounds
.tab-nav / .tab-pane	Tab switching via site.js initTabs()
.badge / .badge-amber	Category pills (news, tags)

7A.4 Motion Classes (flagship-motion.js)

Class	Entrance
.fx.fx-rise	Translate Y + fade
.fx.fx-scale	Scale from 0.92 + fade
.fx.fx-left / .fx-right	Horizontal slide + fade
.fx.fx-clip	Clip-path reveal (images, media)
.fx.fx-mask	Mask wipe for headlines
.fx.fx-wipe	Horizontal rule wipe for eyebrows/dividers
.fx.in	Added by IntersectionObserver when visible

Stagger via --fx-d custom property (seconds). Scroll-linked emphasis via data-fx-scroll → --fxp (0 to 1).

7A.5 theme.css Legacy Equivalents

Legacy (theme)	Meridian (flagship)
.reveal / .reveal.visible	.fx / .fx.in
.section-dark	.fs-on-dark
.divider-yellow	.fs-rule
.hero-stat	.fs-stat
--yellow	--gold
--radius 4px	--radius 0

11A. Assistant — Function Reference

Function	Description
t(key)	Resolve assistant.* i18n key with English fallback
lang()	Current language: en, fr, or sw
getIndex(l)	Lazy-build FlexSearch index for language l
search(query, l)	Return ranked document IDs for query
respond(query)	Format top hit as bot message with read-more link
buildUI(mount)	Replace #chat-widget inner DOM with assistant panel
storage.load/add/clear	IndexedDB messages with in-memory fallback
openPanel/closePanel	Toggle .assistant-open on panel + body scroll lock

11A.1 Search Algorithm

1. Strip stopwords (the, a, is, ...) from query
2. Search FlexSearch index with tokenized query
3. Boost documents with f : 1 (curated facts)
4. Prefer longer token matches over substring noise
5. Return top 3 hits; format first as answer with source URL

11A.2 Curated Facts Categories

build_assistant_kb.js embeds hand-written facts for: company stats, HQ address, phone, email, country list, subsidiary names, fleet size, station count, steel capacity, LPG cylinder sizes, founding year, founder name. All sourced from scripts/_verified_lake_facts.md.

11A.3 Offline Behaviour

On 404.html and offline.html, assistant works without i18n-content.js — uses English FALLBACK dict. KB and FlexSearch still load from precache. IndexedDB persists conversation across sessions when available.

19. Troubleshooting Guide

19.1 Translations Not Switching

Symptom: Clicking FR/SW does nothing.

Causes:

- `i18n-content.js` not loaded or loaded after `i18n.js`
- Under `file://` with old build that used `fetch()` — rebuild `i18n-content.js`
- Missing `data-i18n` attribute on element (key not in dictionary shows English fallback — looks like "not working")

Fix: Verify script order in `<body>`. Check console for "`LAKE_I18N_CONTENT__` is not defined". Run `node scripts/_qa_check_i18n.js`.

19.2 3D Hero Black Box

Symptom: Empty or black `#fuel-experience` section.

Causes:

- `hero-3d.bundle.js` not built or 404
- WebGL unsupported or disabled
- JavaScript error in bundle — check console
- Lazy load IO never triggered — scroll to section

Fix: Rebuild bundle. Hard-refresh. Check Network tab for bundle load. Under reduced motion, static finale is expected.

19.3 Service Worker Serving Stale Content

Symptom: Deployed changes not visible.

Fix: Bump `VERSION` in `sw.js`. User must accept update toast or hard-refresh twice. `news-data.js` and `hero-3d.bundle.js` use `network-first` specifically to avoid this.

19.4 Africa Map Shows Circles Not Borders

Cause: data_countries_africa.js not loaded before africa-network-map.js, or
__LAKE_AFRICA_GEOJSON__ undefined.

Fix: Verify script order. Check console for GeoJSON error.

19.5 Assistant Returns "Couldn't Find That" for Known Facts

Fix: Rebuild KB: node scripts/build_assistant_kb.js. Ensure i18n keys exist for the content. Try exact terms: "Lake Oil", "Tanzania", "LPG", "contact".

19.6 Nav Active Link Wrong

Cause: Previously substring-matched hrefs. Fixed in site.js to compare exact filenames.

Fix: Ensure href="fuel.html" not href="/fuel" without extension if comparing logic expects filename.

20. Firebase Hosting Configuration

package.json defines:

```
{
  "scripts": {
    "serve": "npx firebase emulators:start --only hosting",
    "deploy": "npx firebase deploy --only hosting"
  },
  "devDependencies": {
    "firebase-tools": "^13.35.1"
  }
}
```

A typical firebase.json (create locally) serves the repo root as public directory:

```
{
  "hosting": {
    "public": ".",
    "ignore": ["firebase.json", "**/.*", "**/node_modules/**", "lake-3d/**", "scripts/_chrome_profile*/**"],
    "headers": [
      {
        "source": "/sw.js",
        "headers": [{ "key": "Cache-Control", "value": "no-cache" }]
      },
      {
        "source": "**/*.@(js|css)",
        "headers": [{ "key": "Cache-Control", "value": "public,max-age=3600" }]
      }
    ],
    "rewrites": [
      { "source": "/", "destination": "/index.html" }
    ]
  }
}
```

Important: sw.js must be served with no-cache or short TTL so updates propagate. Production domain: www.lakeoilgroup.com.

21. Image Asset Conventions

Directory	Contents	SW Strategy
assets/images/banner/	Division hero banners (LakeOil, LakeGas, etc.)	cache-first
assets/images/news/{id}/	News article photos	cache-first
assets/images/n-slider/	Homepage news slider	stale-while-revalidate
assets/images/planet/	Earth textures for 3D hero	cache-first
assets/images/leadership/	Executive photos, events	cache-first
assets/images/flags/	Country SVG flags (8 ops countries)	cache-first
lake-story-assets/	our-story.html slideshow scenes	stale-while-revalidate
assets/images/og-cover.jpg	Default Open Graph image	cache-first

Use descriptive alt text. Add data-i18n-alt for translated descriptions. Prefer JPEG for photos, PNG for logos with transparency, SVG for icons and flags.

22. i18n Key Naming Conventions

Prefix	Scope	Example
nav.*	Navigation labels	nav.home, nav.fuel, nav.dd.energy
mob.*	Mobile nav	mob.services
footer.*	Footer columns	footer.contact, footer.privacy
chat.*	Legacy chat + assistant UI	chat.placeholder, chat.reply.fuel
assistant.*	Knowledge assistant	assistant.greeting, assistant.nomatch
hero.*	Homepage hero	hero.title, hero.sub
fuel.*	fuel.html content	fuel.hero.title, fuel.spec.1
about.*	about.html content	about.story.p1
stat.*	Homepage statistics	stat.employees, stat.trucks

Page prefix matches filename stem (underscore for hyphens: container_services.*). Numeric suffixes for list items. Keep keys stable — changing keys requires full rebuild.

23. Extended Page Documentation

23.1 404.html — Full Developer Notes

Standard Meridian page for 404 content. See Section 5 for summary.

Document Structure

- Lines of HTML: **173**
- Title tag: Page Not Found | Lake Group
- Meta description: (none)
- Section IDs: none
- H2 headings: varies
- JSON-LD types: none
- 3D hero: no

Asset Dependencies

CSS: `assets/flagship.css`, `assets/assistant.css`

JS: `assets/pwa.js` → `assets/flagship-motion.js` →
`assets/vendor/flexsearch/flexsearch.bundle.min.js` → `assets/assistant-kb.js` →
`assets/assistant.js`

Content Editing Workflow

1. Edit HTML content; add `data-i18n="pagekey.element"` to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects `prefers-reduced-motion`
- ☐ Images have alt text
- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.2 about.html — Full Developer Notes

Standard Meridian page for about content. See Section 5 for summary.

Document Structure

- Lines of HTML: **585**
- Title tag: About Lake Group | Our Story & Vision
- Meta description: Founded in 2006 by Ally Edha Awadh, Lake Group has grown from a single fuel outlet into one of East Africa's largest industrial conglomerates.
- Section IDs: our-story-embed
- H2 headings: It started | to a fleet of | we brought clean | We built more than | across | Lake Group. | From a Single Fuel Company to a Regional Powerhouse | What Drives Us | Where We're Going | A Word from Our Founder
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text

- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.3 africa-network.html — Full Developer Notes

Most JavaScript-heavy page after index. Requires live network for Esri/OSM tiles. Country cards call `window.selectCountry(id, cardElement)` which invokes `LakeAfricaMap.flyToCountry()`. Legend buttons filter asset types. GeoJSON borders highlight 8 operations countries in gold fill.

Document Structure

- Lines of HTML: **571**
- Title tag: Africa Network | Lake Group Operations Map
- Meta description: Lake Group operations across Tanzania, Kenya, Zambia, Rwanda, Burundi, DRC, Ethiopia, Mozambique and Dubai.
- Section IDs: none
- H2 headings: Spread Across Every Corner of East & Central Africa | Why Lake Group Stands Out Across Africa | Operations Across the Continent
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: `assets/flagship.css`, `assets/assistant.css`,
`assets/vendor/leaflet/leaflet.css`

JS: `assets/vendor/leaflet/leaflet.js` → `assets/data_countries_africa.js` →
`assets/africa-network-map.js` → `assets/i18n-content.js` → `assets/i18n.js` →
`assets/site.js` → `assets/vendor/flexsearch/flexsearch.bundle.min.js` →
`assets/assistant-kb.js` → `assets/assistant.js` → `assets/pwa.js` → `assets/flagship-motion.js`

Content Editing Workflow

1. Edit HTML content; add `data-i18n="pagekey.element"` to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px

- ☐ No horizontal overflow (node scripts/_qa_overflow_check.js)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.4 careers.html — Full Developer Notes

Standard Meridian page for careers content. See Section 5 for summary.

Document Structure

- Lines of HTML: **288**
- Title tag: Careers at Lake Group | Jobs in East Africa
- Meta description: Join over 4,600 professionals across 21 nationalities at Lake Group. Explore career opportunities across our 20+ subsidiaries.
- Section IDs: apply
- H2 headings: Build Your Career Across Africa | Drop Us Your CV
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.5 concrete.html — Full Developer Notes

Standard Meridian page for concrete content. See Section 5 for summary.

Document Structure

- Lines of HTML: **311**
- Title tag: Concrete & Aggregate | GCCP | Lake Group
- Meta description: Gulf Concrete & Cement Products — Dar es Salaam's leading ready-mix concrete and aggregate supplier.
- Section IDs: none
- H2 headings: Building East Africa's Future | Our Plant in Action | Specialist Concrete Portfolio
- JSON-LD types: Service, Organization, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: `assets/flagship.css`, `assets/assistant.css`

JS: `assets/i18n-content.js` → `assets/i18n.js` → `assets/site.js` → `assets/vendor/flexsearch/flexsearch.bundle.min.js` → `assets/assistant-kb.js` → `assets/assistant.js` → `assets/pwa.js` → `assets/flagship-motion.js`

Content Editing Workflow

1. Edit HTML content; add `data-i18n="pagekey.element"` to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects `prefers-reduced-motion`
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.6 contact.html — Full Developer Notes

Standard Meridian page for contact content. See Section 5 for summary.

Document Structure

- Lines of HTML: **287**
- Title tag: Contact Lake Group | Get in Touch
- Meta description: Contact Lake Group headquarters in Dar es Salaam. Phone, email and location details for all regional offices.
- Section IDs: none
- H2 headings: We're Happy to Hear From You
- JSON-LD types: ContactPage, Organization, ContactPoint, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.7 container-services.html — Full Developer Notes

Standard Meridian page for container services content. See Section 5 for summary.

Document Structure

- Lines of HTML: **277**
- Title tag: Container Services | AFICD & ACFS | Lake Group
- Meta description: Inland container depot and freight station services across Tanzania, Zambia and Mozambique.
- Section IDs: none
- H2 headings: The Inland Gateway to East African Trade
- JSON-LD types: Service, Organization, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.8 csr.html — Full Developer Notes

Standard Meridian page for csr content. See Section 5 for summary.

Document Structure

- Lines of HTML: **277**
- Title tag: CSR & Sustainability | Lake Group
- Meta description: Lake Group's commitment to community development, environmental responsibility and sustainable business across Africa.
- Section IDs: none
- H2 headings: Responsible Growth. Lasting Impact. | How We Give Back
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.9 dashboard.html — Full Developer Notes

Standard Meridian page for dashboard content. See Section 5 for summary.

Document Structure

- Lines of HTML: **337**
- Title tag: Client Dashboard | Lake Group
- Meta description: Lake Group client portal for tracking shipments and managing your account.
- Section IDs: none
- H2 headings: Sign In to Your Account | Welcome back,
- JSON-LD types: none
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.10 fleet.html — Full Developer Notes

Standard Meridian page for fleet content. See Section 5 for summary.

Document Structure

- Lines of HTML: **291**
- Title tag: Our Fleet | Lake Trans | Lake Group
- Meta description: 700+ trucks operating bulk liquid haulage and logistics across East and Central Africa.
- Section IDs: none
- H2 headings: Purpose-Built for Every Load
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.11 fuel.html — Full Developer Notes

Primary product page for Lake Oil petroleum operations. Covers retail network (85+ stations), bulk supply, import infrastructure, and competitive positioning. Photography from `assets/images/banner/LakeOil.jpg` and operational photos. Tab components may separate retail vs bulk content — wired via `site.js initTabs()`.

Document Structure

- Lines of HTML: **284**
- Title tag: Fuel & Petroleum | Lake Group
- Meta description: Lake Oil — Top 5 petroleum distributor in Tanzania with 85+ retail stations and bulk supply across 8 countries.
- Section IDs: none
- H2 headings: Keeping Africa Moving | Our Competitive Edge
- JSON-LD types: Service, Organization, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: `assets/flagship.css`, `assets/assistant.css`

JS: `assets/i18n-content.js` → `assets/i18n.js` → `assets/site.js` → `assets/vendor/flexsearch/flexsearch.bundle.min.js` → `assets/assistant-kb.js` → `assets/assistant.js` → `assets/pwa.js` → `assets/flagship-motion.js`

Content Editing Workflow

1. Edit HTML content; add `data-i18n="pagekey.element"` to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects `prefers-reduced-motion`

- ☐ Images have alt text
- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.12 gallery.html — Full Developer Notes

Masonry layout (.gallery-masonry) with lightbox (#lightbox). Click handlers on .photo-card open full-size image. motion.js skip zone includes #lightbox.

Document Structure

- Lines of HTML: **442**
- Title tag: Gallery | Lake Group
- Meta description: Photos from Lake Group operations, events, and facilities across East and Central Africa.
- Section IDs: none
- H2 headings: Lake Group in Action
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.13 history.html — Full Developer Notes

Standard Meridian page for history content. See Section 5 for summary.

Document Structure

- Lines of HTML: **285**
- Title tag: Our History | Lake Group Since 2006
- Meta description: The story of Lake Group: from a single fuel outlet in Dar es Salaam in 2006 to a pan-African conglomerate.
- Section IDs: none
- H2 headings: Two Decades of Growth | Be Part of the Next Chapter
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.14 index.html — Full Developer Notes

The homepage is the most complex page in the site. It contains ~2900 lines including a large inline `<style>` block that duplicates and extends design tokens for first-paint performance before `theme.css` loads. The hero area above the fold uses a static background image with headline; the 3D experience is a separate full-viewport section lower on the page (`#fuel-experience`).

The 3D section includes a copy column (`.experience-steps`) with three bullet steps that sync to the globe chapters via `syncSteps()`. A DOM finale overlay (`.experience-finale`) shows the Lake Group logo and tagline when `gp > 0.88`.

Division cards link to all 8 service verticals. Stats use `data-count` attributes animated by `site.js`. News teaser may link to `news.html`. Founder section includes portrait from `assets/images/resource/ceo.jpg`.

Document Structure

- Lines of HTML: **2931**
- Title tag: Lake Group | Powering East & Central Africa
- Meta description: Lake Group — East and Central Africa's fastest-growing energy, logistics and industrial conglomerate. Fuel, LPG, steel, concrete and transport across 8 countries.
- Section IDs: `fuel-experience`
- H2 headings: One Network. | A Conglomerate Built on Grit and Vision | Eight Sectors. One Vision. | See Lake Group in Action | A Word from Our Founder | Ready to Work Together?
- JSON-LD types: Organization, Person, PostalAddress
- 3D hero: yes

Asset Dependencies

CSS: `assets/theme.css`, `assets/assistant.css`

JS: `assets/i18n-content.js` → `assets/i18n.js` → `assets/site.js` → `assets/vendor/flexsearch/flexsearch.bundle.min.js` → `assets/assistant-kb.js` → `assets/assistant.js` → `assets/pwa.js` → `assets/motion.js`

Content Editing Workflow

1. Edit HTML content; add `data-i18n="pagekey.element"` to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.

4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node scripts/_qa_overflow_check.js)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.15 investors.html — Full Developer Notes

Includes currency selector (#currency-select) wired by site.js initCurrency(). Elements with data-invest-usd hold USD base values converted to TZS/KES/ZMW on change. Financial figures should match verified facts document.

Document Structure

- Lines of HTML: **269**
- Title tag: Investor Relations | Lake Group
- Meta description: Financial performance, governance and investment information for Lake Group — one of East Africa's fastest-growing conglomerates.
- Section IDs: none
- H2 headings: A Track Record of Vertical Growth | What Drives Our Value
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text

- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.16 leadership.html — Full Developer Notes

Standard Meridian page for leadership content. See Section 5 for summary.

Document Structure

- Lines of HTML: **291**
- Title tag: Leadership | Lake Group
- Meta description: Meet the executive team leading Lake Group across energy, logistics, steel, concrete and LPG operations in East Africa.
- Section IDs: none
- H2 headings: Guided by Experience. Driven by Vision. | Ally Edha Awadh | Group & Lake Oil Management | Companies Under Lake Group
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.17 logistics.html — Full Developer Notes

Standard Meridian page for logistics content. See Section 5 for summary.

Document Structure

- Lines of HTML: **301**
- Title tag: Transport & Haulage | Lake Trans | Lake Group
- Meta description: Bulk liquid and dry cargo haulage with 700+ trucks across Tanzania, Kenya, Zambia and the wider region.
- Section IDs: none
- H2 headings: The Backbone of East Africa's Supply Chain | Purpose-Built for Africa's Roads
- JSON-LD types: Service, Organization, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.18 lpg.html — Full Developer Notes

Standard Meridian page for lpg content. See Section 5 for summary.

Document Structure

- Lines of HTML: **290**
- Title tag: LPG Gas | Lake Gas | Lake Group
- Meta description: LPG bottling, composite cylinders and distribution across East and Central Africa.
- Section IDs: none
- H2 headings: Clean Energy for Every Kitchen | Lake Gas Across Africa
- JSON-LD types: Service, Organization, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.19 lubricants.html — Full Developer Notes

Standard Meridian page for lubricants content. See Section 5 for summary.

Document Structure

- Lines of HTML: **267**
- Title tag: Lubricants | Lake Lubes | Lake Group
- Meta description: Lake Lubes manufactures quality lubricants and greases for automotive and industrial applications.
- Section IDs: none
- H2 headings: Engineered for Performance
- JSON-LD types: Service, Organization, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.20 media-center.html — Full Developer Notes

Standard Meridian page for media center content. See Section 5 for summary.

Document Structure

- Lines of HTML: **272**
- Title tag: Media Center | Lake Group
- Meta description: Press releases, media kits and corporate communications from Lake Group.
- Section IDs: none
- H2 headings: Latest Announcements
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.21 news-article.html — Full Developer Notes

Standard Meridian page for news article content. See Section 5 for summary.

Document Structure

- Lines of HTML: **243**
- Title tag: News Article | Lake Group
- Meta description: Read the full story from Lake Group news and events.
- Section IDs: none
- H2 headings: varies
- JSON-LD types: BreadcrumbList, ListItem, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/news-data.js → assets/news.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.22 news.html — Full Developer Notes

Standard Meridian page for news content. See Section 5 for summary.

Document Structure

- Lines of HTML: **300**
- Title tag: News & Events | Lake Group
- Meta description: Latest announcements, expansions and community activities from Lake Group across Africa.
- Section IDs: none
- H2 headings: Recent News
- JSON-LD types: CollectionPage, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/news-data.js → assets/news.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.23 offline.html — Full Developer Notes

Standard Meridian page for offline content. See Section 5 for summary.

Document Structure

- Lines of HTML: **215**
- Title tag: You're Offline | Lake Group
- Meta description: (none)
- Section IDs: none
- H2 headings: varies
- JSON-LD types: none
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/pwa.js → assets/flagship-motion.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js

Content Editing Workflow

1. Edit HTML content; add `data-i18n="pagekey.element"` to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects `prefers-reduced-motion`
- ☐ Images have alt text
- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.24 our-story.html — Full Developer Notes

Cinematic slideshow not in main nav. Eight scenes from lake-story-assets/scene1-8.jpg. Auto-advance timer plus click/tap/Space/ArrowRight to advance. Full viewport .ose-stage. Can be embedded in about.html via #our-story-embed iframe.

Document Structure

- Lines of HTML: **619**
- Title tag: Lake Group: Our Story
- Meta description: The Lake Group story — from one fuel depot in Dar es Salaam in 2006 to an energy, logistics and industrial group operating across 8 African countries. An immersive scroll experience.
- Section IDs: none
- H2 headings: It started | to a fleet of | we brought clean | We built more than | across | Lake Group. | "From a single fuel outlet to one of East Africa's
- JSON-LD types: none
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/pwa.js → assets/flagship-motion.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run python3 scripts/build_master_en.py to extract new keys.
3. Add FR/SW translations in scripts/translation_dict.py.
4. Run python3 scripts/build_i18n_content.py.
5. Run node scripts/build_assistant_kb.js to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node scripts/_qa_overflow_check.js)
- ☐ All links valid

- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.25 projects.html — Full Developer Notes

Standard Meridian page for projects content. See Section 5 for summary.

Document Structure

- Lines of HTML: **249**
- Title tag: Major Projects | Lake Group
- Meta description: Major construction, logistics and industrial projects delivered by Lake Group subsidiaries.
- Section IDs: none
- H2 headings: Delivering East Africa's Development
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.26 services.html — Full Developer Notes

Standard Meridian page for services content. See Section 5 for summary.

Document Structure

- Lines of HTML: **375**
- Title tag: Services | Lake Group
- Meta description: Lake Group operates across 8 business sectors: fuel & petroleum, LPG gas, transport & haulage, concrete, lubricants, steel, container services and more across East Africa.
- Section IDs: none
- H2 headings: One Integrated Vision | Ready to Work Together?
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js → assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js → assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.27 station-locator.html — Full Developer Notes

Standard Meridian page for station locator content. See Section 5 for summary.

Document Structure

- Lines of HTML: **310**
- Title tag: Station Locator | Lake Oil | Lake Group
- Meta description: Find your nearest Lake Oil fuel station across Tanzania and the region.
- Section IDs: none
- H2 headings: 85+ Stations Across Tanzania
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct
- ☐ Page works offline after first visit (SW precache)

23.28 steel.html — Full Developer Notes

Standard Meridian page for steel content. See Section 5 for summary.

Document Structure

- Lines of HTML: **295**
- Title tag: Lake Steel | HS-CR Rebars | Lake Group
- Meta description: Tanzania's first HS-CR rebar manufacturer with 100,000 MT annual rolling mill capacity.
- Section IDs: none
- H2 headings: First in Tanzania. Built to Last. | From Billets to Bars
- JSON-LD types: Service, Organization, BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
4. Run `python3 scripts/build_i18n_content.py`.
5. Run `node scripts/build_assistant_kb.js` to update assistant.
6. Test language switch and assistant queries for changed content.

QA Checklist

- ☐ Responsive at 375px, 768px, 1024px, 1440px
- ☐ No horizontal overflow (node `scripts/_qa_overflow_check.js`)
- ☐ All links valid
- ☐ Motion respects prefers-reduced-motion
- ☐ Images have alt text
- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

23.29 sustainability.html — Full Developer Notes

Standard Meridian page for sustainability content. See Section 5 for summary.

Document Structure

- Lines of HTML: **251**
- Title tag: Sustainability | Lake Group
- Meta description: Environmental stewardship, clean cooking LPG access and sustainable operations across Lake Group.
- Section IDs: none
- H2 headings: Energy for Africa. Responsibly. | Sustainability by the Numbers
- JSON-LD types: BreadcrumbList, ListItem, ListItem
- 3D hero: no

Asset Dependencies

CSS: assets/flagship.css, assets/assistant.css

JS: assets/i18n-content.js → assets/i18n.js → assets/site.js →
assets/vendor/flexsearch/flexsearch.bundle.min.js → assets/assistant-kb.js →
assets/assistant.js → assets/pwa.js → assets/flagship-motion.js

Content Editing Workflow

1. Edit HTML content; add data-i18n="pagekey.element" to translatable nodes.
2. Run `python3 scripts/build_master_en.py` to extract new keys.
3. Add FR/SW translations in `scripts/translation_dict.py`.
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- ☐ Canonical URL correct

- ☐ Page works offline after first visit (SW precache)

24. JavaScript — Extended Module Documentation

24.x assets/site.js

initNav()

Binds #nav-toggle click to toggle .open on #nav-mobile. Sets display:flex/none inline as fallback if CSS fails. Marks active links by comparing window.location.pathname filename to href filename (strips query/hash).

initReveal()

Immediately shows .reveal elements already in viewport. Creates IntersectionObserver with threshold 0.08 and rootMargin bottom -40px. Adds .visible class on intersect; unobserves after reveal.

initCounters()

Parses data-count (integer), data-prefix, data-suffix. Animates over 1600ms with ease-out cubic. Fallback: sets final value after 2500ms if observer never fired. Sets data-animated=1 to prevent re-run.

initChat()

Legacy keyword chat. Guarded by __LAKE_ASSISTANT_ACTIVE__. Uses word-boundary RegExp matching; prefers longest matching keyword. Replies via Lakel18n.t('chat.reply.'+key) with English fallbacks.

initCurrency()

investors.html only. Rates object: USD=1, TZS=2650, KES=153, ZMW=27.5. Formats billions/millions for USD display.

24.x assets/i18n.js

Public API

```
window.LakeI18n = {  
  init(),           // bind lang buttons, apply stored lang  
  apply(lang),      // switch language, persist, re-render  
  t(key, lang?),    // lookup translation or null  
  get current()     // 'en' | 'fr' | 'sw'  
}
```

Events

Dispatches lake-i18n-applied CustomEvent with detail.lang after each apply. hero-3d.js and assistant.js listen for this to refresh localized strings.

24.x assets/sw.js

Self-contained service worker — no imports. VERSION bump triggers cache purge on activate. trimCache() implements LRU-ish eviction on image cache (max 150). Auth query params never cached. Cross-origin requests pass through unhandled.

24.x assets/news.js

getArticleId()

Parses ?id= or ?newsid= from URL. Returns 0 if missing.

renderArticle()

Builds article DOM: header, banner img, description paragraphs, image grid, YouTube iframe if video URL present. Updates <title> and meta description. Injects NewsArticle schema.

filterNews()

Client-side filter on LAKE_NEWS array. Re-renders #news-list. Re-calls LakeSite.initReveal() for new cards.

24.x assets/pwa.js

IIFE, no exports. Resolves SW URL via new URL('../sw.js', document.currentScript.src). Toast is imperative DOM (not in template) for isolation from i18n. reloadingAfterUpdate flag prevents reload loops.

assets/i18n-content.js

Build artifact: window.__LAKE_I18N_CONTENT__ = { en: {...}, fr: {...}, sw: {...} }. ~1,442 keys. Generated by scripts/build_i18n_content.py from scripts/_master_en.json + translation_dict.py.

See Section 6 for summary. Source in repository assets/ directory.

assets/motion.js

Pairs with theme.css. Adds html.lg-motion. Auto-tags .reveal on grids/orphans. Nav hide/show on scroll. Card 3D tilt (pointer:fine). Respects prefers-reduced-motion.

See Section 6 for summary. Source in repository assets/ directory.

assets/flagship-motion.js

Pairs with flagship.css. Adds html.fs-motion. Richer fx-* choreography (fx-rise, fx-clip, fx-mask, fx-wipe). Scroll progress via [data-fx-scroll]. Magnetic CTA hover. [data-fs-count] counters.

See Section 6 for summary. Source in repository assets/ directory.

assets/hero-3d.js

Three.js module (bundled to hero-3d.bundle.js). See Section 9.

See Section 6 for summary. Source in repository assets/ directory.

assets/hero-3d.bundle.js

esbuild output: hero-3d.js + three.module.min.js as classic script. ~0.5MB. Rebuild via scripts/build_hero_bundle.sh.

See Section 6 for summary. Source in repository assets/ directory.

assets/assistant.js

Offline knowledge assistant. FlexSearch retrieval over assistant-kb.js. IndexedDB lake-assistant/messages. Sets __LAKE_ASSISTANT_ACTIVE__. UI replaces #chat-widget.

See Section 6 for summary. Source in repository assets/ directory.

assets/assistant-kb.js

Build artifact: window.__LAKE_ASSISTANT_KB__. Generated by scripts/build_assistant_kb.js.

See Section 6 for summary. Source in repository assets/ directory.

assets/news-data.js

window.LAKE_NEWS array — article metadata, images, categories, dates.

See Section 6 for summary. Source in repository assets/ directory.

assets/africa-network-map.js

Leaflet map init. ASSETS, PIPELINES, COUNTRY_META. window.LakeAfricaMap API.
GeoJSON from __LAKE_AFRICA_GEOJSON__.

See Section 6 for summary. Source in repository assets/ directory.

assets/data_countries_africa.js

window.__LAKE_AFRICA_GEOJSON__ — Africa country borders for offline/file:// map.

See Section 6 for summary. Source in repository assets/ directory.

25. Common Developer Workflows

25.1 Add a New Static Page

1. Copy about.html as template (flagship migrated)
2. Update title, meta, canonical, JSON-LD
3. Replace main content; add data-i18n keys
4. Add to scripts/templates/nav.html if needed
5. Run normalize_nav.py
6. Add to sitemap.xml
7. Add PAGES entry in build_assistant_kb.js
8. Run full i18n + KB build pipeline
9. Add URL to sw.js PRECACHE if linked from offline.html
10. Deploy and verify

25.2 Update Translations

1. Edit scripts/translation_dict.py (PHRASES_FR, PHRASES_SW, TERMS_FR, TERMS_SW)
2. python3 scripts/build_master_en.py
3. python3 scripts/build_i18n_content.py
4. node scripts/build_assistant_kb.js
5. Test FR and SW on affected pages

25.3 Update 3D Globe Sites

1. Verify facts in scripts/_verified_lake_facts.md
2. Edit SITES and/or FACILITIES in hero-3d.js
3. Update SITE_NAMES for en/fr/sw
4. bash scripts/build_hero_bundle.sh
5. Bump ?v= on index.html
6. node scripts/_globe_qa2.js

25.4 Roll Out Flagship to index.html

1. Replace theme.css link with flagship.css
2. Replace motion.js with flagship-motion.js
3. Refactor inline CSS to use flagship tokens
4. Retag .reveal elements as .fx where needed
5. Visual QA entire homepage
6. Update README migration status

26. Security & Privacy Notes

- No user data sent to server — static site, mock forms only
- Assistant conversations stored locally in IndexedDB only
- SW never caches URLs with auth-like query params (token, apikey, etc.)
- No third-party analytics scripts in production HTML
- Map tiles contact Esri/OSM servers — privacy policy should disclose
- dashboard.html is demo UI only — disallow in robots.txt

27. Performance Budget

Asset	Approx Size	Load Strategy
index.html inline CSS	~80KB	First paint critical
flagship.css	~45KB	Render-blocking link
hero-3d.bundle.js	~500KB	Lazy on IO
i18n-content.js	~200KB	defer
assistant-kb.js	~150KB	defer
Font files (woff2)	~300KB total	preload critical

Target LCP: hero image/text before 3D loads. 3D is below fold — must not block. motion.js uses transform/opacity only to avoid layout thrash.

28. Version History & Migration Status

Component	Status	Notes
Flagship CSS rollout	28/29 pages	index.html remains on theme.css
i18n full-site architecture	Complete	Partial translation coverage
PWA / Service Worker	v10	Bump VERSION per deploy
Knowledge Assistant	Live	Replaces legacy chat
hero-3d bundle	v9 query param	file:// safe
lake-3d Next.js	Orphaned	Not in production
Portuguese (PT)	Removed	Replaced with Swahili (SW)

29. Complete News Article Index

All articles in `assets/news-data.js` (20 total). Images under `assets/images/news/{id}/`.

ID	Title	Date	Category	URL
20	Lake Group New Year 2016	01 Jan, 2019	Events	news-article.html?id=20
19	Lake Group Football Team	01 Jan, 2019	Sports	news-article.html?id=19
18	Lake Group Annual Event	01 Jan, 2019	Events	news-article.html?id=18
17	Car Rally	01 Jan, 2019	Events	news-article.html?id=17
16	Auto Fest 10th	01 Jan, 2019	Events	news-article.html?id=16
15	2018 Lake Group Event	01 Jan, 2019	Events	news-article.html?id=15
14	Corporate Social Responsibilities	01 Jan, 2019	CSR	news-article.html?id=14
13	Composite cylinder launch on 18th June, 2014	18 Jun, 2014	LPG	news-article.html?id=13
12	Indian Food Festival event held on 23rd Feb, 2014	01 Jan, 2019	Events	news-article.html?id=12
11	Africa Food Festival held on 30th November, 2014	01 Jan, 2019	Events	news-article.html?id=11
10	International Women's day 2015	01 Jan, 2015	CSR	news-article.html?id=10
9	International Women's Day 2014	01 Jan, 2019	CSR	news-article.html?id=9
8	Lake Gas Advertisement	01 Jan, 2019	LPG	news-article.html?id=8
7	Lake Group Expansion into Mozambique Market	01 Jan, 2019	Expansion	news-article.html?id=7
6	East Africa's biggest LPG storage facility in Tanga, Tanzania	01 Jan, 2019	LPG	news-article.html?id=6
5	Donation of Mattress to Temeke hospital in DAR	01 Jan, 2019	CSR	news-article.html?id=5

ID	Title	Date	Category	URL
4	International Women's Day (IWD) 2017 In Tanzania	01 Jan, 2019	CSR	news-article.html? id=4
3	MOU signed with Moroccan King	01 Jan, 2019	Business	news-article.html? id=3
2	AFRICA CEO FORUM 2018 AWARDS	01 Jan, 2019	Awards	news-article.html? id=2
1	MOU signed with SINO Trucks	01 Jan, 2019	Logistics	news-article.html? id=1

30. Lake Group Subsidiaries & Page Mapping

Lake Group operates through 20+ subsidiaries across energy, logistics, and industrial sectors. The website maps each major division to a dedicated service page.

Subsidiary	Sector	Page	Key Facts
Lake Oil	Fuel & Petroleum	fuel.html	Top 5 distributor Tanzania; 85+ stations; 8 countries
Lake Gas	LPG	lpg.html	6kg–38kg cylinders; composite cylinders; Tanga 3,000 MT terminal
Lake Lubes	Lubricants	lubricants.html	Automotive and industrial greases
Lake Steel	Steel	steel.html	First HS-CR rebar in Tanzania; 100,000 MT/yr; Kibaha mill
GCCP	Concrete	concrete.html	Ready-mix Dar es Salaam; Lugoba quarry; est. 2010
Lake Trans	Logistics	logistics.html , fleet.html	700+ trucks; bulk liquid and dry cargo
AFICD / ACFS	Containers	container-services.html	Inland depots Tanzania, Zambia, Mozambique
Burundi Petroleum	Fuel	africa-network.html	Bujumbura operations
Lake Petroleum (RW, ZM)	Fuel	africa-network.html	Kigali, Ndola/Lusaka
Wadi Elsundus / WAS	Fuel	africa-network.html	Ethiopia — Addis Ababa, Gelan depot
MERM / SAFF	Concrete / Trading	africa-network.html	Dubai UAE operations

31. Service Worker Precache Manifest

URLs installed on SW activate (36 entries from sw.js PRECACHE_URLS):

URL
./index.html
./offline.html
./404.html
./about.html
./services.html
./africa-network.html
./contact.html
./manifest.webmanifest
./assets/pwa.js
./assets/site.js
./assets/theme.css
./assets/motion.js
./assets/flagship.css
./assets/flagship-motion.js
./assets/i18n.js
./assets/i18n-content.js
./assets/assistant.js
./assets/assistant.css
./assets/assistant-kb.js
./assets/vendor/flexsearch/flexsearch.bundle.min.js
./assets/images/logos/LAKE_GROUP_LOGO.jpg
./assets/icons/pwa/icon-192.png
./assets/icons/pwa/icon-512.png
./assets/fonts/fonts.css
./assets/fonts/files/bebas-neue-latin-400-normal.woff2

URL
./assets/fonts/files/inter-latin-300-normal.woff2
./assets/fonts/files/inter-latin-400-normal.woff2
./assets/fonts/files/inter-latin-500-normal.woff2
./assets/fonts/files/inter-latin-600-normal.woff2
./assets/fonts/files/inter-latin-700-normal.woff2
./assets/fonts/files/inter-latin-800-normal.woff2
./assets/fonts/files/playfair-display-latin-400-normal.woff2
./assets/fonts/files/playfair-display-latin-400-italic.woff2
./assets/fonts/files/playfair-display-latin-500-normal.woff2
./assets/fonts/files/playfair-display-latin-600-normal.woff2
./assets/fonts/files/material-symbols-outlined-latin-400-normal.woff2

32. Frequently Asked Questions

Q1. Why two CSS files?

The site is migrating from theme.css to flagship.css (Meridian design). Only index.html still uses the legacy pair.

Q2. Why i18n-content.js instead of JSON?

Browsers block fetch() of local JSON under file://. Script tags work under file://, http://, and https://.

Q3. Why bundle hero-3d.js?

ES module scripts are blocked under file://. esbuild produces a classic script with Three.js inlined.

Q4. How do I add a language?

Extend SUPPORTED in i18n.js, add dictionary in build pipeline, update lang switcher in nav template, rebuild all i18n artifacts.

Q5. Does the assistant use AI?

No. It retrieves verbatim passages from assistant-kb.js via FlexSearch. No LLM, no network calls.

Q6. Why is dashboard.html blocked?

It is a mock demo portal, not a real authenticated application. robots.txt Disallow prevents indexing.

Q7. What is lake-3d/?

Orphaned Next.js prototype. Not linked from production. Delete or budget separate migration project.

Q8. How often bump sw VERSION?

Every deploy that changes any precached file. When in doubt, bump.

Q9. Can I use npm packages on the live site?

Avoid runtime CDN deps. Vendor libs live in assets/vendor/. Build step may use npm (esbuild, firebase-tools).

Q10. Where are fonts hosted?

Self-hosted under assets/fonts/files/. fonts.css defines @font-face. No Google Fonts CDN.

Q11. What map tiles does africa-network use?

Esri World Imagery, OpenTopoMap, OpenStreetMap — all require network at runtime.

Q12. How does news filtering work?

Pure client-side on LAKE_NEWS array. Category exact match; country via keyword heuristics in article text.

Q13. What is the Meridian tick?

8×8px gold square on hairline rule — signature flagship motif in eyebrows and section markers.

Q14. Why defer scripts?

Non-blocking parse. Order preserved. DOMContentLoaded fires after deferred scripts execute.

Q15. How to test offline?

Load site online, wait for SW install, DevTools → Network → Offline, navigate to about.html.

33. Structured Data by Page

Page	schema.org Types	Notes
404.html	—	WebPage / Service / BreadcrumbList typical
about.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
africa-network.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
careers.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
concrete.html	Service, Organization, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
contact.html	ContactPage, Organization, ContactPoint, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
container-services.html	Service, Organization, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
csr.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
dashboard.html	—	WebPage / Service / BreadcrumbList typical
fleet.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
fuel.html	Service, Organization, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
gallery.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
history.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
index.html	Organization, Person, PostalAddress	Organization + founder
investors.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
leadership.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical

Page	schema.org Types	Notes
logistics.html	Service, Organization, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
lpg.html	Service, Organization, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
lubricants.html	Service, Organization, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
media-center.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
news-article.html	BreadcrumbList, ListItem, ListItem, ListItem	Dynamic NewsArticle on news-article.html
news.html	CollectionPage, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
offline.html	—	WebPage / Service / BreadcrumbList typical
our-story.html	—	WebPage / Service / BreadcrumbList typical
projects.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
services.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
station-locator.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
steel.html	Service, Organization, BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical
sustainability.html	BreadcrumbList, ListItem, ListItem	WebPage / Service / BreadcrumbList typical

34. Operations Countries — Verified Reference

All geographic claims on the 3D hero and africa-network map must match scripts/_verified_lake_facts.md.

Country	ISO	Base City	Entities	Map Key
Tanzania	TZ	Dar es Salaam (HQ)	Lake Oil, Lake Gas, Lake Steel, GCCP, Lake Trans, AFICD	tz
Kenya	KE	Nairobi	Lake Oil Kenya, Mombasa port access	ke
Zambia	ZM	Lusaka / Ndola	Lake Petroleum Zambia, AFICD	zm
Rwanda	RW	Kigali	Lake Petroleum Rwanda, Lake Gas	rw
Burundi	BI	Bujumbura	Burundi Petroleum	bi
DR Congo	CD	Lubumbashi	DRC Petroleum, Lake Trans	cd
Ethiopia	ET	Addis Ababa	Wadi Elsundus Petroleum	et
Mozambique	MZ	Beira / Maputo	Lake Oil LDA, AFICD	mz
UAE	AE	Dubai	MERM Ready Mix, SAFF	ae

35. Template System Reference

Shared chrome is maintained in four template files under `scripts/templates/`:

35.1 nav.html

Desktop navigation with three dropdown menus (Services, Network, Company), language switcher (EN/FR/SW buttons with `.lang-btn` and `data-lang`), and logo link to `index.html`. All labels use `data-i18n` keys under `nav.*` prefix.

35.2 mobile_nav.html

Accordion-style mobile menu mirroring desktop links. Toggled by `#nav-toggle`. Uses `mob.*` `i18n` keys. Hidden by default; `.open` class shows flex layout.

35.3 footer.html

Four-column footer grid: company links, services links, network links, contact info. Social media icons. Copyright line. `footer.*` `i18n` keys.

35.4 chat_widget.html

Legacy chat markup: `#chat-widget`, `#chat-btn`, `#chat-box`, `#chat-messages`, `#chat-input`. `Assistant.js` replaces inner UI but preserves mount point. `chat.*` `i18n` keys for placeholder and greeting.

35.5 normalize_nav.py

Python script that splices templates into all root HTML files between known markers. Run after any template edit. Prevents nav drift between pages.

36. Architectural Decision Records

36.1 ADR-001: Static MPA over SPA Framework

Context: Corporate site with 29 pages, SEO requirements, offline PWA, and limited dev team.

Decision: Plain HTML pages with shared vanilla JS modules.

Rationale: Zero framework runtime cost. Each page is independently cacheable. No hydration complexity. Works on file:// for local preview. Firebase Hosting serves static files efficiently.

Consequences: Nav/footer duplication solved by template scripts. No client-side router. Page transitions are full reloads (acceptable for corporate site).

36.2 ADR-002: Script-Loaded Globals over fetch() for Critical Data

Context: Stakeholders preview site by double-clicking HTML files.

Decision: Ship i18n, GeoJSON, assistant KB, and 3D bundle as script tags setting window.* globals.

Rationale: file:// CORS blocks fetch() silently. Script tags have no such restriction.

36.3 ADR-003: FlexSearch Assistant over External Chatbot

Context: Need helpful Q&A without API costs, privacy concerns, or hallucination risk.

Decision: Build-time KB + client-side search returning verbatim passages.

Rationale: Works offline. Answers are auditable. No API keys. Multilingual via parallel indexes.

36.4 ADR-004: Dual Design System During Migration

Context: Full visual redesign (Meridian) cannot ship atomically for 29 pages.

Decision: flagship.css pair replaces theme.css pair per-page. Both precached until complete.

Rationale: Incremental rollout reduces risk. index.html last due to 3D section coupling.

36.5 ADR-005: Lazy-Load 3D Bundle

Context: hero-3d.bundle.js is ~500KB.

Decision: IntersectionObserver on #fuel-experience with 600px rootMargin.

Rationale: 3D is below fold. LCP should not wait for WebGL. Users who never scroll still get full page.

37. flagship.css Class Index (Sample)

flagship.css contains 1139 lines. Selected class selectors:

Class
.alert
.alert-info
.alert-success
.badge
.badge-blue
.badge-green
.badge-red
.badge-yellow
.breadcrumb
.btn
.btn-blue
.btn-lg
.btn-outline
.btn-outline-dark
.btn-primary
.btn-sm
.btn-white
.card
.card-body
.card-icon
.container
.country-cell
.country-tag
.d-flex
.dd-label

Class
.div-card
.divider-yellow
.division-dot
.division-item
.ev-row
.ev-title
.flag-icon
.flag-icon-lg
.footer-bottom
.footer-bottom-links
.footer-brand
.footer-col
.footer-grid
.footer-logo
.footer-motto
.footer-social
.form-group
.form-row
.fs-band
.fs-card-no
.fs-container-narrow
.fs-container-wide
.fs-corners
.fs-display
.fs-eyebrow

Class
.fs-head
.fs-head-center
.fs-hero-meta
.fs-hero-title
.fs-index
.fs-lede
.fs-marker
.fs-media
.fs-media-caption
.fs-on-dark
.fs-outline
.fs-outline-gold
.fs-page-light
.fs-rule
.fs-section
.fs-serif-quote
.fs-split
.fs-split-even
.fs-split-rev
.fs-stat
.fs-stat-rail
.fs-surface-2
.fs-table
.fw-600
.fx

Class
.g-item
.grid-2
.grid-3
.grid-4
.has-dropdown
.lang-btn
.lang-switcher
.leader-card
.leader-photo
.marquee-item
.marquee-wrap
.mb-8
.mob-section
.msg
.msg-bot
.msg-user
.mt-8
.nav-cta
.nav-dropdown
.nav-inner
.nav-links
.nav-logo
.nav-mobile
.nav-toggle
.news-item

Class
<code>.news-read-more</code>
<code>.page-hero</code>
<code>.page-wrapper</code>
<code>.photo-card</code>
<code>.prj-card</code>
<code>.reveal</code>
<code>.section</code>
<code>.section-dark</code>
<code>.section-label</code>
<code>.section-light</code>
<code>.section-sm</code>
<code>.section-subtitle</code>
<code>.section-title</code>
<code>.side-link</code>
<code>.site-footer</code>
<code>.site-nav</code>
<code>.social-link</code>
<code>.sus-emoji</code>
<code>.sus-item</code>
<code>.tab-nav</code>

See `FLAGSHIP_DESIGN.md` for complete component documentation.

38. Document Information

This developer guide is designed for print at A4 with 2cm margins. Estimated page count depends on browser print settings; content density targets 80+ pages when printed from Chrome or Firefox with default margins.

To regenerate after repository changes:

```
python3 docs/_gen_developer_guide.py
```

Source of truth is always the repository files — this document is generated from live project metadata and should be regenerated before major releases.

39. Page-by-Page Technical Essays

404.html

This document describes the technical characteristics of 404.html, a production page in the Lake Group static website.

Purpose: Corporate content page for 404.

The page comprises approximately 173 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared `flagship.css` (or `theme.css` on `index.html`) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: standard section headers. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: `assets/flagship.css`, `assets/assistant.css`. JavaScript executes in dependency order: `assets/pwa.js`, `assets/flagship-motion.js`, `assets/vendor/flexsearch/flexsearch.bundle.min.js`, `assets/assistant-kb.js`, `assets/assistant.js`.

Internationalization uses `data-i18n` attributes keyed by page prefix. After content edits, developers must run `build_master_en.py` and `build_i18n_content.py` to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: No JSON-LD on this page. Pages with `BreadcrumbList` schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

about.html

About.html serves dual roles: it is the primary company overview for visitors and the reference implementation for the Meridian flagship design system. Every other migrated page follows patterns established here — fs-hero-title in the page hero, fs-on-dark ink bands alternating with paper sections, fs-marker numbered section headers, and fs-split asymmetric layouts.

The page embeds the cinematic our-story experience via #our-story-embed, allowing visitors to watch the eight-scene brand film without leaving the about flow. The narrative sections walk through founding in 2006, fleet growth, LPG introduction, steel and concrete diversification, and pan-African expansion. Photography is drawn from verified operational assets — storage terminals, trucks, leadership events.

Values and mission content uses data-i18n keys under the about.* prefix. When French or Swahili is selected, translated strings replace text; untranslated long-form paragraphs gracefully remain in English rather than showing raw keys. JSON-LD typically includes WebPage and BreadcrumbList types for SEO.

Developers editing this page should treat it as the canonical template: copy structure, not just styles, when creating new pages. After edits, run the full i18n pipeline and rebuild the assistant knowledge base so the about.* chunks stay current.

africa-network.html

This document describes the technical characteristics of africa-network.html, a production page in the Lake Group static website.

Purpose: Lake Group operations across Tanzania, Kenya, Zambia, Rwanda, Burundi, DRC, Ethiopia, Mozambique and Dubai.

The page comprises approximately 571 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Spread Across Every Corner of East & Central Africa, Why Lake Group Stands Out Across Africa, Operations Across the Continent. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css, assets/vendor/leaflet/leaflet.css. JavaScript executes in dependency order: assets/vendor/leaflet/leaflet.js, assets/data_countries_africa.js, assets/africa-network-map.js, assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

careers.html

This document describes the technical characteristics of careers.html, a production page in the Lake Group static website.

Purpose: Join over 4,600 professionals across 21 nationalities at Lake Group. Explore career opportunities across our 20+ subsidiaries.

The page comprises approximately 288 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Build Your Career Across Africa, Drop Us Your CV. Section IDs present: apply.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

concrete.html

This document describes the technical characteristics of concrete.html, a production page in the Lake Group static website.

Purpose: Gulf Concrete & Cement Products — Dar es Salaam's leading ready-mix concrete and aggregate supplier.

The page comprises approximately 311 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Building East Africa's Future, Our Plant in Action, Specialist Concrete Portfolio. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types Service, Organization, BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

contact.html

This document describes the technical characteristics of contact.html, a production page in the Lake Group static website.

Purpose: Contact Lake Group headquarters in Dar es Salaam. Phone, email and location details for all regional offices.

The page comprises approximately 287 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: We're Happy to Hear From You. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types ContactPage, Organization, ContactPoint, BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

container-services.html

This document describes the technical characteristics of container-services.html, a production page in the Lake Group static website.

Purpose: Inland container depot and freight station services across Tanzania, Zambia and Mozambique.

The page comprises approximately 277 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: The Inland Gateway to East African Trade. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types Service, Organization, BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

csr.html

This document describes the technical characteristics of csr.html, a production page in the Lake Group static website.

Purpose: Lake Group's commitment to community development, environmental responsibility and sustainable business across Africa.

The page comprises approximately 277 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Responsible Growth. Lasting Impact., How We Give Back.
Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

dashboard.html

This document describes the technical characteristics of dashboard.html, a production page in the Lake Group static website.

Purpose: Lake Group client portal for tracking shipments and managing your account.

The page comprises approximately 337 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Sign In to Your Account, Welcome back,. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: No JSON-LD on this page. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

fleet.html

This document describes the technical characteristics of fleet.html, a production page in the Lake Group static website.

Purpose: 700+ trucks operating bulk liquid haulage and logistics across East and Central Africa.

The page comprises approximately 291 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Purpose-Built for Every Load. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

fuel.html

Fuel.html represents Lake Oil, the group's flagship petroleum division and the business from which Lake Group grew. Content covers the retail network of 85+ stations across Tanzania, bulk supply to industrial customers, import terminal infrastructure, and competitive advantages in the East African fuel market.

The page uses standard Meridian layout: immersive page hero with Bebas display type, alternating ink and paper sections, photography from assets/images/banner/LakeOil.jpg and operational depot images. Tab components, if present, are wired through site.js initTabs() — clicking tab buttons toggles .active on .tab-pane elements within the nearest .tab-container.

All user-facing strings should carry data-i18n keys prefixed fuel.*. Product specifications, station counts, and country lists must align with scripts/_verified_lake_facts.md — the same document feeding the 3D globe and assistant curated facts. Schema.org markup typically declares Service type with provider Organization.

Performance-wise, fuel.html loads only the standard deferred script stack — no map libraries, no 3D, no news modules. This keeps the page lightweight for users on 3G connections searching for fuel station information.

gallery.html

This document describes the technical characteristics of gallery.html, a production page in the Lake Group static website.

Purpose: Photos from Lake Group operations, events, and facilities across East and Central Africa.

The page comprises approximately 442 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Lake Group in Action. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

history.html

This document describes the technical characteristics of history.html, a production page in the Lake Group static website.

Purpose: The story of Lake Group: from a single fuel outlet in Dar es Salaam in 2006 to a pan-African conglomerate.

The page comprises approximately 285 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Two Decades of Growth, Be Part of the Next Chapter. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

index.html

The homepage is the technical centerpiece of the Lake Group website. Unlike every other page, it retains the legacy theme.css design system because the #fuel-experience section was engineered alongside the original inline styles and the experience-3d DOM structure. Migrating index.html to Meridian (flagship.css) is the final step of the design rollout and requires careful regression testing of the 3D globe section, which has explicit skip zones in both motion engines.

The page structure follows a deliberate performance hierarchy. The first screen is a static hero with headline, subcopy, and call-to-action buttons — no WebGL, no heavy scripts. This ensures Largest Contentful Paint remains fast on mobile networks common in East Africa. The 3D interactive globe sits in #fuel-experience, typically one to two viewport heights below the fold. An inline IntersectionObserver script (not in a separate file) loads hero-3d.bundle.js only when the section approaches the viewport, with a generous 600px rootMargin so the bundle begins downloading before the user arrives.

The 3D section pairs WebGL canvas with a DOM copy column (.experience-steps) listing three narrative beats that sync to globe chapters. A finale overlay (.experience-finale) fades in during Chapter 4 of each 39-second loop, displaying the Lake Group logo and tagline over the living globe. Users who prefer reduced motion see this finale statically without WebGL initialization.

Below the 3D section, the divisions grid links to all eight service verticals. Statistics use data-count attributes consumed by site.js initCounters(). The founder section establishes corporate credibility with portrait photography. A news teaser may surface recent announcements. The closing CTA band drives contact and careers conversions.

Script load order on index.html: i18n-content, i18n, site, flexsearch, assistant-kb, assistant, inline lazy 3D loader, pwa, motion. Notably absent from initial load: hero-3d.bundle.js (lazy), Leaflet, news-data.js. Organization JSON-LD in head provides rich search results for company name, founder, address, and social profiles.

investors.html

This document describes the technical characteristics of investors.html, a production page in the Lake Group static website.

Purpose: Financial performance, governance and investment information for Lake Group — one of East Africa's fastest-growing conglomerates.

The page comprises approximately 269 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: A Track Record of Vertical Growth, What Drives Our Value.
Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

leadership.html

This document describes the technical characteristics of leadership.html, a production page in the Lake Group static website.

Purpose: Meet the executive team leading Lake Group across energy, logistics, steel, concrete and LPG operations in East Africa.

The page comprises approximately 291 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Guided by Experience. Driven by Vision., Ally Edha Awadh, Group & Lake Oil Management, Companies Under Lake Group. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

logistics.html

This document describes the technical characteristics of logistics.html, a production page in the Lake Group static website.

Purpose: Bulk liquid and dry cargo haulage with 700+ trucks across Tanzania, Kenya, Zambia and the wider region.

The page comprises approximately 301 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: The Backbone of East Africa's Supply Chain, Purpose-Built for Africa's Roads. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types Service, Organization, BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

lpg.html

This document describes the technical characteristics of lpg.html, a production page in the Lake Group static website.

Purpose: LPG bottling, composite cylinders and distribution across East and Central Africa.

The page comprises approximately 290 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Clean Energy for Every Kitchen, Lake Gas Across Africa. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types Service, Organization, BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

lubricants.html

This document describes the technical characteristics of lubricants.html, a production page in the Lake Group static website.

Purpose: Lake Lubes manufactures quality lubricants and greases for automotive and industrial applications.

The page comprises approximately 267 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Engineered for Performance. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types Service, Organization, BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

media-center.html

This document describes the technical characteristics of media-center.html, a production page in the Lake Group static website.

Purpose: Press releases, media kits and corporate communications from Lake Group.

The page comprises approximately 272 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Latest Announcements. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

news-article.html

This document describes the technical characteristics of news-article.html, a production page in the Lake Group static website.

Purpose: Read the full story from Lake Group news and events.

The page comprises approximately 243 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: standard section headers. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/news-data.js, assets/news.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

news.html

This document describes the technical characteristics of news.html, a production page in the Lake Group static website.

Purpose: Latest announcements, expansions and community activities from Lake Group across Africa.

The page comprises approximately 300 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Recent News. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/news-data.js, assets/news.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types CollectionPage, BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

offline.html

This document describes the technical characteristics of offline.html, a production page in the Lake Group static website.

Purpose: Corporate content page for Offline.

The page comprises approximately 215 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: standard section headers. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/pwa.js, assets/flagship-motion.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: No JSON-LD on this page. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

our-story.html

This document describes the technical characteristics of our-story.html, a production page in the Lake Group static website.

Purpose: The Lake Group story — from one fuel depot in Dar es Salaam in 2006 to an energy, logistics and industrial group operating across 8 African countries. An immersive scroll experience.

The page comprises approximately 619 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: It started, to a fleet of, we brought clean, We built more than, across. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/pwa.js, assets/flagship-motion.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: No JSON-LD on this page. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

projects.html

This document describes the technical characteristics of projects.html, a production page in the Lake Group static website.

Purpose: Major construction, logistics and industrial projects delivered by Lake Group subsidiaries.

The page comprises approximately 249 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Delivering East Africa's Development. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

services.html

This document describes the technical characteristics of services.html, a production page in the Lake Group static website.

Purpose: Lake Group operates across 8 business sectors: fuel & petroleum, LPG gas, transport & haulage, concrete, lubricants, steel, container services and more across East Africa.

The page comprises approximately 375 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: One Integrated Vision, Ready to Work Together?. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

station-locator.html

This document describes the technical characteristics of station-locator.html, a production page in the Lake Group static website.

Purpose: Find your nearest Lake Oil fuel station across Tanzania and the region.

The page comprises approximately 310 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: 85+ Stations Across Tanzania. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

steel.html

This document describes the technical characteristics of steel.html, a production page in the Lake Group static website.

Purpose: Tanzania's first HS-CR rebar manufacturer with 100,000 MT annual rolling mill capacity.

The page comprises approximately 295 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: First in Tanzania. Built to Last., From Billets to Bars. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types Service, Organization, BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

sustainability.html

This document describes the technical characteristics of sustainability.html, a production page in the Lake Group static website.

Purpose: Environmental stewardship, clean cooking LPG access and sustainable operations across Lake Group.

The page comprises approximately 251 lines of HTML including inline critical CSS in the document head. This inline CSS pattern is consistent across all Lake Group pages — it establishes layout and page-specific overrides before the shared flagship.css (or theme.css on index.html) loads. The inline block should not duplicate tokens already defined in the shared stylesheet except where page-specific layout demands it.

Structural headings include: Energy for Africa. Responsibly., Sustainability by the Numbers. Section IDs present: none declared.

Asset loading follows the site-wide convention. CSS: assets/flagship.css, assets/assistant.css. JavaScript executes in dependency order: assets/i18n-content.js, assets/i18n.js, assets/site.js, assets/vendor/flexsearch/flexsearch.bundle.min.js, assets/assistant-kb.js, assets/assistant.js, assets/pwa.js, assets/flagship-motion.js.

Internationalization uses data-i18n attributes keyed by page prefix. After content edits, developers must run build_master_en.py and build_i18n_content.py to regenerate dictionaries. The assistant knowledge base should also be rebuilt so queries about this page return current text.

Structured data: Types BreadcrumbList, ListItem, ListItem. Pages with BreadcrumbList schema help search engines understand site hierarchy.

For QA, verify responsive layout at mobile (375px), tablet (768px), and desktop (1440px) breakpoints. Confirm no horizontal overflow. Test language switching on navigation labels and any tagged body content. Verify offline availability after service worker precache on second visit.

This page does not load the 3D hero or Leaflet map libraries. Standard Meridian design applies.

40. Build Scripts Encyclopedia

scripts/_add_ose_keys.js

Type: .js · **Lines:** 100

Utility script in scripts/. Review source for usage.

Invocation: `node scripts/_add_ose_keys.js` from repository root.

scripts/_apply_audit_fixes.js

Type: .js · **Lines:** 464

QA/automation utility: _apply_audit_fixes.js. Run manually during release testing.

Invocation: `node scripts/_apply_audit_fixes.js` from repository root.

scripts/_assistant_qa.js

Type: .js · **Lines:** 354

Utility script in scripts/. Review source for usage.

Invocation: `node scripts/_assistant_qa.js` from repository root.

scripts/_globe_qa.js

Type: .js · **Lines:** 284

QA/automation utility: _globe_qa.js. Run manually during release testing.

Invocation: `node scripts/_globe_qa.js` from repository root.

scripts/_globe_qa2.js

Type: .js · **Lines:** 286

QA/automation utility: _globe_qa2.js. Run manually during release testing.

Invocation: `node scripts/_globe_qa2.js` from repository root.

scripts/_qa_check_i18n.js

Type: .js · **Lines:** 42

QA/automation utility: _qa_check_i18n.js. Run manually during release testing.

Invocation: `node scripts/_qa_check_i18n.js` from repository root.

scripts/_qa_check_links.js

Type: .js · **Lines:** 65

QA/automation utility: _qa_check_links.js. Run manually during release testing.

Invocation: `node scripts/_qa_check_links.js` from repository root.

scripts/_qa_matrix.js

Type: .js · **Lines:** 86

QA/automation utility: _qa_matrix.js. Run manually during release testing.

Invocation: `node scripts/_qa_matrix.js` from repository root.

scripts/_qa_overflow_check.js

Type: .js · **Lines:** 212

QA/automation utility: _qa_overflow_check.js. Run manually during release testing.

Invocation: `node scripts/_qa_overflow_check.js` from repository root.

scripts/add_assistant_tags.js

Type: .js · **Lines:** 84

Batch-injects assistant.css, flexsearch, assistant-kb.js, assistant.js tags.

Invocation: `node scripts/add_assistant_tags.js` from repository root.

scripts/add_pwa_tags.js

Type: .js · **Lines:** 72

Batch-injects manifest link, theme-color meta, apple-touch-icon into HTML heads.

Invocation: `node scripts/add_pwa_tags.js` from repository root.

scripts/add_seo_tags.js

Type: .js · **Lines:** 277

Batch-injects description, og:*, twitter:card, canonical link tags.

Invocation: `node scripts/add_seo_tags.js` from repository root.

scripts/add_theme_motion_tags.js

Type: .js · **Lines:** 66

Batch-injects theme.css/flagship.css and motion.js/flagship-motion.js link tags.

Invocation: `node scripts/add_theme_motion_tags.js` from repository root.

scripts/build_assistant_kb.js

Type: .js · **Lines:** 495

Node script generating assets/assistant-kb.js from i18n-content.json plus CURATED_FACTS.
Chunks page content into ~340 char documents for FlexSearch.

Invocation: `node scripts/build_assistant_kb.js` from repository root.

scripts/build_hero_bundle.sh

Type: .sh · **Lines:** 16

Shell wrapper invoking esbuild to bundle hero-3d.js with three.module.min.js into hero-3d.bundle.js as IIFE classic script.

Invocation: `bash scripts/build_hero_bundle.sh` from repository root.

scripts/build_i18n_content.py

Type: .py · **Lines:** 100

Reads _master_en.json and translation_dict.py output to generate assets/i18n-content.json and assets/i18n-content.js. The .js file wraps JSON in window.__LAKE_I18N_CONTENT__ assignment for file:// compatibility.

Invocation: `python3 scripts/build_i18n_content.py` from repository root.

scripts/build_master_en.py

Type: .py · **Lines:** 119

Scans all HTML files for data-i18n attributes and merges with existing _master_en.json to produce the English master dictionary. Run before adding translations.

Invocation: `python3 scripts/build_master_en.py` from repository root.

scripts/build_sw_lang.js

Type: .js · **Lines:** 109

Generates language-specific service worker precache variants for multilingual offline shells.

Invocation: `node scripts/build_sw_lang.js` from repository root.

scripts/fix_i18n_script_tag.py

Type: .py · **Lines:** 50

One-off fix script for i18n script tag ordering issues.

Invocation: `python3 scripts/fix_i18n_script_tag.py` from repository root.

scripts/generate_dev_pdf.sh

Type: .sh · **Lines:** 18

Helper to render documentation to PDF via headless browser.

Invocation: `bash scripts/generate_dev_pdf.sh` from repository root.

scripts/i18n_extract.py

Type: .py · **Lines:** 336

Discovers i18n keys in HTML, reports orphans and missing keys. Use when auditing translation coverage.

Invocation: `python3 scripts/i18n_extract.py` from repository root.

scripts/inline_material_icons.py

Type: .py · **Lines:** 94

Inlines Material Symbols as SVG or self-hosted font for offline use.

Invocation: `python3 scripts/inline_material_icons.py` from repository root.

scripts/normalize_nav.py

Type: .py · **Lines:** 136

Splices nav.html, mobile_nav.html, footer.html, chat_widget.html into all root HTML pages.
Critical for chrome consistency.

Invocation: `python3 scripts/normalize_nav.py` from repository root.

scripts/perf_pass.js

Type: .js · **Lines:** 101

Automated performance audit — checks script defer, preload hints, image dimensions.

Invocation: `node scripts/perf_pass.js` from repository root.

scripts/self_host_fonts.py

Type: .py · **Lines:** 57

Downloads font files from fontsource or similar and generates assets/fonts/fonts.css @font-face rules.

Invocation: `python3 scripts/self_host_fonts.py` from repository root.

scripts/translation_dict.py

Type: .py · **Lines:** 2340

Source of truth for French and Swahili translations. Contains PHRASES_FR, PHRASES_SW (full sentences) and TERMS_FR, TERMS_SW (short labels). Imported by build_i18n_content.py.

Invocation: `python3 scripts/translation_dict.py` from repository root.